

Active-Workload Equity in Electric Vehicle Routing with Time Windows and Partial Recharging: A Multi-objective Fleet-Prioritized Pareto Adaptive Large Neighborhood Search

Abstract

Electric vehicle routing with time windows and partial recharging requires routing, charging, and service schedules to be coordinated jointly, yet routing efficiency does not necessarily imply an equitable distribution of operational duties. This study introduces active-workload equity into the electric vehicle routing problem with time windows and partial recharging (E-VRPTW-PR). Active workload is defined as the time spent traveling, serving customers, and charging, while time-window-induced waiting is kept separate in route completion time. This distinction allows workload equity and schedule exposure to be evaluated as related but different operational criteria. We formulate a fleet-first multi-objective problem that prioritizes the number of utilized vehicles and subsequently considers travel distance, active-workload inequality measured by the Gini coefficient, and maximum route completion time in the Pareto sense. To solve the problem, we develop a fleet-prioritized Pareto adaptive large neighborhood search integrating partial-recharging evaluation, adaptive destroy–repair mechanisms, local-search intensification, and multi-objective archive and acceptance control. Computational experiments on 56 benchmark instances show that explicit equity guidance produces lower-inequality alternatives and broader equity-inclusive trade-off sets at common fleet levels while retaining competitive fleet sizes. Distance-focused solutions require only a small median distance premium of approximately 0.4%, and the equity advantage becomes pronounced when modest distance flexibility is allowed, particularly from a 3% distance budget onward, without evidence of a systematic change in maximum route completion time. The results show that active-workload equity is distinct from route-duration balance and that Pareto-based planning can expose practically useful compromises among routing efficiency, workload distribution, and schedule performance.

Keywords: Electric vehicle routing; Time windows; Partial recharging; Workload equity; Multi-objective optimization; Adaptive large neighborhood search

1 Introduction

Workload equity and balanced use of operational resources are becoming increasingly important in real-world logistics. Logistics decisions are not driven solely by transportation cost; for instance, highly uneven workloads can create additional organizational and economic consequences (Nekooghadirli et al. 2026). More balanced operations may help reduce overtime exposure, improve employee satisfaction and retention, support more consistent customer service, and enable more flexible use of available fleet capacity. These considerations are particularly relevant in vehicle-routing applications, where operational efficiency and workforce-related outcomes are closely connected.

While electric vehicles have been increasingly promoted as a means of reducing emissions in response to evolving regulations and transportation restructuring (Santero et al. 2026; Nguyen et al. 2022), their integration into routing decisions introduces additional interdependencies among route design, charging decisions, and service scheduling. A change in customer assignment may propagate through the remaining route by altering downstream service times, charging requirements, and energy feasibility. Workload allocation in this setting therefore cannot be characterized solely by the number of assigned customers or the distance traveled. Routes with similar travel distances may involve substantially different amounts of travel, customer service, and charging time. In addition, route completion time may include waiting induced by customer time windows, which contributes to schedule duration without representing an active operational task. A meaningful assessment of workload distribution consequently requires active operational duties to be distinguished from overall route completion time.

Previous vehicle-routing studies have examined workload equity (Nekooghadirli et al. 2026; Mahmoodian et al. 2025), while the electric vehicle routing literature has extensively considered time-window and partial-recharging decisions (Wang and Zhao 2023; Xiao et al. 2026; Yu and Anh 2026). Nevertheless, limited attention has been devoted to the distribution of active workload when time-window-induced waiting and partial-recharging operations are considered jointly. This study addresses this gap by incorporating active-workload equity into the Electric Vehicle Routing Problem with Time Windows and Partial Recharging (E-VRPTW-PR). The analysis examines whether explicit equity guidance can reduce active-workload inequality at a common fleet level without disproportionate increases in travel distance or maximum route completion time. The main contributions are summarized as follows.

- Under E-VRPTW-PR context, this paper uncovers active operational workload from route completion time and incorporate that into our models. We show that workload equity can be achieved with a very limited impact on fleet size and travel distance.
- We introduce an efficient multi-objective fleet-prioritized Pareto Adaptive Large Neighborhood Search (ALNS) to solve E-VRPTW-PR.
- At common fleet levels, explicit equity guidance consistently expands the attainable distance–equity decision space and produces lower-inequality alternatives. Distance-focused representatives incur a median travel-distance premium of approximately

0.4%, while budgeted analyses show increasingly pronounced equity improvements from a 3% distance allowance onward, without a systematic change in maximum route completion time. Route-level diagnostics further confirm that lower Gini values are generally accompanied by reduced workload dispersion and lower concentration on the most heavily loaded route. The results indicate that modest routing flexibility can generate meaningful workload-equity gains while preserving fleet requirements and schedule performance.

The remainder of the paper is organized as follows. Section 2 reviews related literature, while Section 3 formally defines our problem setting. Section 4 presents our multi-objective ALNS framework. Section 5 discusses the computational results of the proposed solution and its meaning. Section 6 concludes our paper.

2 Literature Review

Electric vehicle routing problems have long incorporated battery limitations and charging operations into routing decisions (Pelletier et al. 2016; Kucukoglu et al. 2021). Early models commonly assumed full battery recharging at charging stations (Schneider et al. 2014; Desaulniers et al. 2016), whereas later studies introduced partial-recharging strategies that allow vehicles to replenish only the energy required for subsequent operations (Keskin and Catay 2016; Wang and Zhao 2023; Xiao et al. 2026). The literature has since expanded to consider more realistic operational features, including mixed fleets (Goeke and Schneider 2015; Macrina et al. 2019; Wang and Zhao 2023), nonlinear charging processes (Bruni et al. 2025; Froger et al. 2024), battery degradation (Montoya et al. 2024), and time-window constraints (Wang and Zhao 2023; Xiao et al. 2026; Yu and Anh 2026). These developments have substantially improved the operational realism of electric vehicle routing models.

Beyond routing feasibility and transportation efficiency, however, the equitable distribution of workload is also important for the practical deployment of logistics solutions, with potential implications for safety, workforce utilization, and long-term operational sustainability (Nekooghadirli et al. 2026; Mahmoodian et al. 2025; Mancini et al. 2021). At the same time, logistics decision makers often need to balance multiple and potentially conflicting considerations, such as transportation cost, service performance, workload distribution, and resource utilization. Such trade-offs are central to the broader literature on multi-objective vehicle routing (Jozefowiez et al. 2008; Molina et al. 2014; Li 2024; Moghdani et al. 2025).

These considerations become particularly important in electric vehicle routing with time windows and partial recharging. In this setting, total route duration may obscure substantial differences in the actual operational duties assigned to individual routes. A relatively short route, for example, may involve substantial charging activity, whereas a longer route may contain considerable waiting imposed by customer time windows. Moreover, charging and waiting can interact: charging performed before visiting a customer may absorb time that would otherwise be spent waiting for the customer’s time window to open, allowing the service schedule to remain feasible despite an additional station visit. Charging decisions therefore affect not only energy feasibility and route duration but also the composition and distribution of operational workload

across routes. Consequently, although partial recharging, workload equity, and multi-objective routing have each been investigated separately, their combined effects on the distribution of operational duties remain underexplored.

These gaps motivate the present E-VRPTW-PR, in which active workload comprises travel, customer-service, and charging time, whereas route completion time additionally includes waiting induced by customer time windows. We develop a fleet-prioritized Pareto ALNS that first seeks to reduce the number of utilized vehicles and then explores trade-offs among travel distance, active-workload inequality, and maximum route completion time. ALNS is well suited to this constrained setting because its destroy and repair operations can restructure multiple routes while reconsidering charging decisions (Cai et al. 2025; Liu et al. 2024; Ropke and Pisinger 2006; Vidal et al. 2013; Hong et al. 2026). The proposed framework combines Pareto archive management with cyclic search directions inspired by MOEA/D (Zhang and Li 2007). Equity is measured by the Gini coefficient (Matl et al. 2018, 2019) and interpreted jointly with the other operational criteria because the measure is relative and non-monotonic (Karsu and Morton 2015). Section 3 next formalizes the problem and its fleet-first evaluation criterion.

3 Problem Description and Formulation

The problem is defined on a directed graph $G = (N, A)$, in which $N = C \cup S \cup D$ symbolizes the customers C , physical charging locations S , and depots $D = \{0, n+1\}$, with node 0 as the departure depot and node $n+1$ as the return depot. Each customer $i \in C$ is associated with demand q_i , service duration sv_i , and a hard time window $[e_i, l_i]$. A vehicle arriving before e_i may wait until service can begin, whereas arrival after l_i excludes customer i service. Finally, the set of arcs $A = \{(i, j) \mid i, j \in V, j \neq 0, i \neq n+1\}$ represents connections between nodes. For each arc $(i, j) \in A$, d_{ij} , t_{ij} , and hd_{ij} denote the corresponding travel distance, travel time, and energy consumption, respectively.

A homogeneous fleet K is initially located at the departure depot. Each utilized vehicle serves a subset of customers, may visit charging stations multiple times when necessary, and ultimately returns to node $n+1$. Vehicles are subject to a payload capacity $Q_{\max}$ and a battery capacity $B_{\max}$. Battery energy can be replenished at any charging station $s \in S$ at rate g_s . Recharging is partial, so the amount of energy acquired at a station can be adapted to the vehicle’s current battery level, the energy required for subsequent travel, the remaining battery capacity, and the available schedule slack. Note that the same charging station may be visited multiple times. Table 1 summarizes the notation for our problem.

Accordingly, a solution specifies, for each utilized vehicle, an ordered sequence of customer and charging-station visits together with the amount of energy acquired at each charging stop. Feasibility requires that every customer be visited exactly once and served within its time window, while each route continuously satisfies both payload and battery constraints and returns to the depot with sufficient energy to complete all traveled arcs.

Table 1 Notation and description.

Notation	Description
<i>Sets and indices</i>	
$G = (V, A)$	Directed graph representing the expanded transportation network used in the formulation.
C	Set of customer nodes.
S	Set of physical charging-station locations.
D	Set of depot nodes, $D = \{0, n + 1\}$, where 0 and $n + 1$ are the departure and return depots, respectively.
R_s	Finite set of virtual visit copies associated with physical charging station $s \in S$.
S'	Set of virtual charging-station nodes, $S' = \{(s, r) : s \in S, r \in R_s\}$.
$s(i)$	Physical charging-station location associated with virtual node $i \in S'$.
V	Complete node set used in the formulation, $V = C \cup S' \cup D$.
A	Set of feasible directed arcs connecting nodes in V .
K	Total number of available vehicles.
k	Vehicle index, $k = 1, \dots, K$.
<i>Customer and time-window parameters</i>	
q_i	Demand of customer $i \in C$.
sv_i	Service duration at customer $i \in C$; $sv_i = 0$ for $i \in S' \cup D$.
e_i	Earliest allowable service or charging-start time at node $i \in C \cup S'$.
l_i	Latest allowable service-start time at customer $i \in C$ or latest charging-completion time at virtual charging-station node $i \in S'$.
$[e_i, l_i]$	Time window or availability interval associated with node $i \in C \cup S'$.
<i>Travel, payload, and energy parameters</i>	
d_{ij}	Travel distance along arc $(i, j) \in A$.
t_{ij}	Travel time along arc $(i, j) \in A$.
hd_{ij}	Energy consumed when traversing arc $(i, j) \in A$.
$Q_{\max}$	Maximum payload capacity of a vehicle.
$B_{\max}$	Maximum battery capacity of a vehicle.
g_s	Charging rate at physical charging station $s \in S$.
<i>Decision and auxiliary variables</i>	
x_{ij}^k	Binary variable equal to 1 if vehicle k traverses arc $(i, j) \in A$, and 0 otherwise.
u^k	Binary variable equal to 1 if vehicle k is utilized, and 0 otherwise.
y_i^k	Binary variable equal to 1 if vehicle k visits customer or virtual charging-station node $i \in C \cup S'$, and 0 otherwise.
θ_i^k	Start time of service or charging at node i for vehicle k .
L_i^k	Remaining payload of vehicle k after visiting node i .
E_i^k	Battery level of vehicle k upon arrival at node i .
c_i^k	Amount of energy charged by vehicle k at node i .
τ_i^k	Charging duration of vehicle k at node i .
A_{act}^k	Active operational workload of vehicle k .
W^k	Time-window-induced waiting time on the route of vehicle k .
H^k	Completion time of the route operated by vehicle k .
$H^{\max}$	Maximum route completion time among utilized vehicles.

3.1 Problem Formulation

The following formulation specifies the feasibility conditions and objective measures used to evaluate candidate solutions. Because the Gini coefficient depends on the endogenous set of utilized vehicles, the full formulation is nonlinear and is used to specify feasibility and evaluation rather than solved directly as a MILP.

Virtual charging-station representation.

To represent repeated visits to the same physical charging station, each station $s \in S$ is associated with a finite set R_s of visit copies. The set of virtual charging-station nodes is

$$S' = \{(s, r) : s \in S, r \in R_s\}. \quad (1)$$

For each virtual node $i \in S'$, let $s(i)$ denote the corresponding physical station. The complete node set used in the formulation is therefore

$$V = C \cup S' \cup D, \quad D = \{0, n+1\}. \quad (2)$$

Each virtual copy represents one station visit and may be used at most once; multiple copies permit repeated visits to the same physical station.

Decision variables.

The routing decisions are represented by the following binary variables:

$$x_{ij}^k = \begin{cases} 1, & \text{if vehicle } k \text{ traverses arc } (i, j), \\ 0, & \text{otherwise,} \end{cases}$$

for all (i, j) and $k \in K$, and

$$u^k = \begin{cases} 1, & \text{if vehicle } k \text{ is utilized,} \\ 0, & \text{otherwise.} \end{cases}$$

Other related variables are defined as follows:

- θ_i^k : start time of service or charging at node i for vehicle k ;
- L_i^k : remaining payload right after vehicle k visits node i ;
- E_i^k : battery level of vehicle k upon arrival at node i ;
- c_i^k : amount of energy charged by vehicle k at node i ;
- τ_i^k : charging duration of vehicle k at node i ;
- A_{act}^k : active operational workload of vehicle k ;
- H^k : route completion time of vehicle k ;
- W^k : time-window-induced waiting time on the route of vehicle k ;
- $H^{\max}$: maximum completion time among utilized vehicles.

To evaluate the operational characteristics of a route, let A_{act}^k denote the active workload of utilized vehicle k , consisting of travel, customer-service, and charging time. Waiting time W^k is not counted as active workload, but it remains part of the vehicle's overall route completion time H^k .

Variable domains.

The variable domains are

$$\begin{aligned} x_{ij}^k &\in \{0, 1\} & \forall (i, j), \quad \forall k \in K, \\ u^k &\in \{0, 1\} & \forall k \in K, \\ \theta_i^k, L_i^k, E_i^k, c_i^k, \tau_i^k &\geq 0 & \forall i \in V, \quad \forall k \in K, \\ A_{\text{act}}^k, H^k, W^k &\geq 0 & \forall k \in K, \quad H^{\max} \geq 0. \end{aligned}$$

Next, we discuss different constraints related to our problem.

3.1.1 Routing constraints

Each utilized vehicle departs from the departure depot and reaches the return depot exactly once:

$$\sum_{j:(0,j)} x_{0j}^k = u^k \quad \forall k \in K, \quad (3)$$

$$\sum_{i:(i,n+1)} x_{i,n+1}^k = u^k \quad \forall k \in K. \quad (4)$$

Flow conservation is imposed at every customer and virtual station node:

$$\sum_{i:(i,j)} x_{ij}^k = \sum_{\ell:(j,\ell)} x_{j\ell}^k \quad \forall j \in C \cup S', \quad \forall k \in K. \quad (5)$$

Each customer is served exactly once:

$$\sum_{k \in K} \sum_{j:(i,j)} x_{ij}^k = 1 \quad \forall i \in C. \quad (6)$$

For each $i \in C \cup S'$ and $k \in K$, the visit indicator is

$$y_i^k = \sum_{j:(i,j)} x_{ij}^k. \quad (7)$$

Each virtual station copy may be used at most once:

$$\sum_{k \in K} y_i^k \leq 1 \quad \forall i \in S'. \quad (8)$$

Constraint (3) ensures that each vehicle departs the depot exactly once, while constraint (4) implies that all vehicle also return to the depot exactly once at the end of their trips. Constraint (5) maintains the physical connectivity of each vehicle's travel route. Constraint (6) makes sure that each customer is served exactly once. Constraints (7)–(8) together limit the visit to each virtual charging station copy at most one time.

3.1.2 Temporal constraints

All utilized vehicles depart at time zero:

$$\theta_0^k = 0 \quad \forall k \in K. \quad (9)$$

The start time at a visited node must account for the preceding service, charging, and travel durations:

$$\theta_j^k \geq \theta_i^k + sv_i + \tau_i^k + t_{ij} - M(1 - x_{ij}^k) \quad \forall (i, j), \quad \forall k \in K. \quad (10)$$

The node availability intervals are enforced by

$$\theta_i^k \geq e_i - M(1 - y_i^k) \quad \forall i \in C \cup S', \quad \forall k \in K, \quad (11)$$

$$\theta_i^k \leq l_i + M(1 - y_i^k) \quad \forall i \in C, \quad \forall k \in K, \quad (12)$$

$$\theta_i^k + \tau_i^k \leq l_i + M(1 - y_i^k) \quad \forall i \in S', \quad \forall k \in K. \quad (13)$$

For any fixed route and charging plan, the schedule is evaluated using earliest-feasible forward propagation from $\theta_0^k = 0$ from constraint (9). Constraint (10) implies that if a vehicle visits node j after i , then the starting time at j must include the service, charging time, and travel time between i and j . Constraints (11)–(12) together ensure that a vehicle visit a location within that location's time window. In case the visited node represents a charging station, then constraint (13) implies that the charging duration is also taken into account regarding time window constraint.

3.1.3 Payload constraints

Payload propagation is enforced by

$$L_j^k \leq L_i^k - q_j + M(1 - x_{ij}^k) \quad \forall (i, j), \quad \forall k \in K, \quad (14)$$

$$L_j^k \geq L_i^k - q_j - M(1 - x_{ij}^k) \quad \forall (i, j), \quad \forall k \in K. \quad (15)$$

The payload variables satisfy

$$0 \leq L_0^k \leq Q_{\max} u^k \quad \forall k \in K, \quad (16)$$

$$0 \leq L_i^k \leq Q_{\max} y_i^k \quad \forall i \in C \cup S', \quad \forall k \in K. \quad (17)$$

Constraints (14)–(15) are non-binding if vehicle k does not traverse from i to j . If arc (i, j) is used, these two constraints together enforce the payload-balance relation

for vehicle k : the remaining payload after visiting j equals the remaining payload after visiting i minus the demand q_j delivered at j . At the starting depot, constraint (16) implies that payload for vehicle k is positive and bounded by its capacity only when that vehicle is selected for the routing task. Constraint (17) states that at each node i that is not the depot, a vehicle payload at that node is positive and bounded by its capacity only when that node is visited by the focal vehicle.

3.1.4 Battery and charging constraints

Battery propagation along each traversed arc is described by

$$E_j^k \leq E_i^k + c_i^k - hd_{ij} + M(1 - x_{ij}^k) \quad \forall(i, j), \quad \forall k \in K, \quad (18)$$

$$E_j^k \geq E_i^k + c_i^k - hd_{ij} - M(1 - x_{ij}^k) \quad \forall(i, j), \quad \forall k \in K. \quad (19)$$

Every utilized vehicle departs with a full battery:

$$E_0^k = B_{\max} u^k \quad \forall k \in K. \quad (20)$$

Battery levels remain within the vehicle capacity:

$$0 \leq E_i^k \leq B_{\max} y_i^k \quad \forall i \in C \cup S', \quad \forall k \in K, \quad (21)$$

$$0 \leq E_{n+1}^k \leq B_{\max} u^k \quad \forall k \in K. \quad (22)$$

Charging is permitted only at charging-station nodes:

$$c_i^k = 0 \quad \forall i \in C \cup D, \quad \forall k \in K. \quad (23)$$

At a visited charging station, the charging amount is bounded by the available battery headroom:

$$0 \leq c_i^k \leq B_{\max} y_i^k - E_i^k \quad \forall i \in S', \quad \forall k \in K. \quad (24)$$

Constraints (18)–(19) are non-binding if vehicle k does not traverse from i to j . If arc (i, j) is used, these two constraints together enforce the battery-balance relation for vehicle k : the battery level after visiting j equals the battery level after visiting i plus energy recharge c_i^k minus the energy consumption hd_{ij} to travel between these two locations. Constraint (20) suggests that selected vehicles are fully charged before leaving the starting depot. Constraint (21)–(22) state that vehicle k battery level is within battery capacity limit at visited nodes. Constraint (23) exclude infeasible charging locations like customer locations and depots. Constraint (24) permits partial recharging, since a vehicle may leave a charging station at any battery level not exceeding $B_{\max}$.

3.1.5 Active workload and route completion measures

Charging duration:

$$\tau_i^k = \begin{cases} \frac{c_i^k}{g_{s(i)}}, & i \in S', \\ 0, & i \in C \cup D. \end{cases} \quad (25)$$

The active operational workload assigned to vehicle k is

$$A_{\text{act}}^k = \sum_{(i,j)} t_{ij} x_{ij}^k + \sum_{i \in C} sv_i y_i^k + \sum_{i \in S'} \tau_i^k. \quad (26)$$

The completion time of route k is

$$H^k = \theta_{n+1}^k - \theta_0^k. \quad (27)$$

which satisfies

$$0 \leq H^k \leq Mu^k \quad \forall k \in K, \quad (28)$$

Under the earliest-feasible scheduling convention, time-window-induced waiting is

$$W^k = H^k - A_{\text{act}}^k. \quad (29)$$

Constraint (25) presents the functional form of charging duration, which is proportional to the amount of energy acquired and inversely proportional to charging rate. Constraint (26) define active operational workload A_{act}^k of vehicle k as sum of travel, customer-service, and charging time. We retain route completion time H_k as a separate criterion as in constraint (27). This decomposition uses earliest-feasible schedules with no discretionary idling and non-overlapping travel, service, charging, and waiting intervals. Hence, route time outside active operational workload A_{act}^k is time-window-induced waiting as in constraint (29). This choice separates modeled active duties from schedule delay. It also does not treat waiting as guaranteed rest or A_{act}^k as a direct fatigue measure. Finally, constraint (28) links the route completion time H^k to vehicle utilization, ensuring that H^k can take a positive value only when vehicle k is utilized.

3.2 Fleet-first multi-objective evaluation

Regarding our multi-objective problem, we follow a fleet-first multi-objective structure. The primary objective that we consider is to minimize the number of utilized vehicles:

$$\min \quad Z_1 = \sum_{k \in K} u^k. \quad (30)$$

Let $K^+ = \{k \in K : u^k = 1\}$ denote the set of utilized vehicles. Conditional on a fixed fleet size, the following three objectives are minimized in the Pareto sense. The

total distance traveled by the fleet is

$$\min \quad Z_2 = \sum_{k \in K} \sum_{(i,j)} d_{ij} x_{ij}^k. \quad (31)$$

Active-workload inequality is measured by the Gini coefficient:

$$\min \quad Z_3 = \frac{\sum_{k \in K^+} \sum_{\ell \in K^+} |A_{\text{act}}^k - A_{\text{act}}^\ell|}{2|K^+| \sum_{m \in K^+} A_{\text{act}}^m}. \quad (32)$$

The fourth objective is to avoid excessively late route completion by minimizing the longest route duration.

$$\min \quad Z_4 = H^{\max}. \quad (33)$$

such that

$$H^{\max} \geq H^k \quad \forall k \in K. \quad (34)$$

To summarize, the solution is evaluated using a fleet-first criterion: the primary objective Z_1 minimizes the number of utilized vehicles to serve all customers. Then, conditional on the selected fleet size, travel distance Z_2 , active-workload inequality measured by the Gini index Z_3 , and maximum route completion time Z_4 are jointly considered in the Pareto sense. Specifically, a feasible solution s' is preferred to another feasible solution s if $Z_1(s') < Z_1(s)$, or if $Z_1(s') = Z_1(s)$ and s' Pareto dominates s with respect to (Z_2, Z_3, Z_4) . The desired output is therefore a non-dominated set of routing and charging plan at the smallest fleet-size level attained.

4 Solution Methodology: Multi-Objective Fleet-prioritized Pareto Adaptive Large Neighborhood Search (FULL)

We describe our quad-objective framework (FULL) in this section. To address this multi-objective challenge, the proposed solution jointly considers routing, charging, and workload allocation while enforcing time-window, payload, and battery constraints. The solution framework consists of five components, as outlined below:

1. Diverse feasible initialization (Section 4.3);
2. Partial recharging consideration (Section 4.4);
3. Adaptive destroy and repair (Section 4.5–4.6);
4. Local-search intensification (Section 4.7); and
5. Pareto archive management with simulated-annealing acceptance.

We next present the overall procedure in Section 4.1 and discuss its components in the following Sections.

4.1 Overall framework

The preceding overview distinguished the reported Pareto archive from the current search trajectory. This subsection formalizes that separation, which allows FULL to preserve strict fleet-first solution quality in its reported output while retaining sufficient flexibility to explore non-improving intermediate states during the search. At that fleet level, it constructs a diverse approximation of the Pareto trade-off set defined by travel distance Z_2 , active-workload inequality Z_3 , and maximum route completion time Z_4 .

Algorithm 1 presents the overall procedure. The search begins with structurally diverse feasible solutions, which initialize both the current solution s and the Pareto archive A . At each iteration, the algorithm selects a lower-level search direction $\mathbf{w}$ together with an adaptive destroy–repair operator pair. The destroy operator removes part of the current solution to create opportunities for route restructuring, and the repair operator reconstructs a complete plan through direct customer insertions or charging-station-assisted alternatives. Local search may subsequently intensify the reconstructed candidate.

Each candidate is fully re-evaluated because changes in customer assignment, route sequence, or station placement can alter energy requirements and temporal feasibility. This evaluation recomputes charging amounts, route schedules, payload and battery states, active workloads, and all objective values. Only feasible candidates proceed to archive updating and current-solution acceptance.

The archive and the current solution have distinct roles. The archive retains nondominated solutions at the smallest fleet level reached and forms the reported output of the method. The current solution provides the starting point for subsequent neighborhood moves and may temporarily move through inferior or higher-fleet states to support diversification. Operator probabilities are periodically updated from search rewards, cyclic search directions promote exploration across the (Z_2, Z_3, Z_4) trade-off space, and prolonged archive stagnation triggers perturbation. The following subsections describe these mechanisms in detail.

4.2 Pareto archive and candidate acceptance

The proposed framework separates the management of the reported Pareto archive from the acceptance of the current search state. The archive is governed directly by the fleet-first preference relation, whereas current-state acceptance additionally permits controlled transitions through non-improving solutions to support diversification.

Fleet-prioritized Pareto archive.

Let $\mathcal{A}$ denote the current archive, and let

$$Z_1^{\mathcal{A}} = \min_{q \in \mathcal{A}} Z_1(q) \quad (35)$$

denote the best fleet level currently represented in that archive. For each feasible candidate s' , the update rule distinguishes three mutually exclusive cases.

Algorithm 1: Overall fleet-prioritized multi-objective ALNS framework

Input : Problem instance $\mathcal{I}$; stopping criterion; set of lower-level search directions $\mathbf{W}$

Output : Fleet-prioritized Pareto archive A

// Block 1: Initialization

```
1  $\mathcal{S}_0 \leftarrow \text{GENERATEINITIALSOLUTIONS}(\mathcal{I});$ 
2  $s \leftarrow \text{SELECTINITIALCURRENTSOLUTION}(\mathcal{S}_0);$ 
3  $A \leftarrow \text{INITIALIZEARCHIVE}(\mathcal{S}_0);$ 
4 while  $\neg \text{STOPPINGCRITERIONMET}()$  do
    // Block 2: Select search direction and neighborhood operators
5      $\mathbf{w} \leftarrow \text{CURRENTSEARCHDIRECTION}(\mathbf{W});$ 
6      $(d, r) \leftarrow \text{SELECTDESTROYREPAIROPERATORS}();$ 
    // Block 3: Destroy and repair
7      $(s^d, U) \leftarrow d(s);$ 
8      $s' \leftarrow r(s^d, U);$ 
    // Block 4: Optional local-search intensification
9     if  $\text{APPLYLOCALSEARCH}()$  then
10          $s' \leftarrow \text{LOCALSEARCH}(s');$ 
    // Block 5: Partial-recharging evaluation and feasibility verification
11      $s' \leftarrow \text{EVALUATEWITHPARTIALRECHARGING}(s');$ 
12     if  $\text{ISFEASIBLE}(s')$  then
        // Block 6: Archive update and current-state acceptance
13          $(A, \text{status}_A) \leftarrow \text{UPDATEARCHIVE}(A, s');$ 
14          $\text{accepted} \leftarrow \text{ACCEPTCANDIDATE}(s', s, \mathbf{w});$ 
15         if  $\text{accepted}$  then
16              $s \leftarrow s';$ 
17          $\text{UPDATEOPERATORSCORES}(d, r, \text{status}_A, \text{accepted});$ 
    // Block 7: Periodic adaptation and diversification
18     if  $\text{SEGMENTCOMPLETED}()$  then
19          $\text{UPDATEOPERATORPROBABILITIES}();$ 
20          $s \leftarrow \text{SELECTSOLUTIONFORNEXTSEARCHDIRECTION}(A, \mathbf{W});$ 
21     if  $\text{STAGNATIONDETECTED}()$  then
22          $s \leftarrow \text{PERTURB}(A);$ 
23 return  $A;$ 
```

If $Z_1(s') < Z_1^A$, the candidate attains a new best fleet level. The previous archive is discarded and reinitialized as $\mathcal{A} \leftarrow \{s'\}$. If $Z_1(s') = Z_1^A$, the candidate is evaluated against the current archive in the lower-level objective space

$$\mathbf{Z}_L(s) = (Z_2(s), Z_3(s), Z_4(s)).$$

At this common fleet level, s' dominates another solution s if

$$Z_j(s') \leq Z_j(s) \quad \forall j \in \{2, 3, 4\}, \quad \text{and} \quad Z_{j^*}(s') < Z_{j^*}(s) \quad \text{for at least one } j^* \in \{2, 3, 4\}. \quad (36)$$

A same-fleet candidate is inserted into $\mathcal{A}$ only when no archived solution dominates it. Any archive members dominated by the new candidate are then removed. Conversely, if $Z_1(s') > Z_1^A$, the candidate cannot enter the reported archive, although it may still be accepted as the current search state by the diversification mechanism described below.

This update rule preserves strict fleet-size priority in the reported decision set while retaining a nondominated approximation of the (Z_2, Z_3, Z_4) trade-off set at the best fleet level found. When the archive exceeds its capacity, objective duplicates are first removed and a diversity-preserving truncation procedure is applied in the normalized (Z_2, Z_3, Z_4) space. The archive capacity, duplicate tolerance, and truncation settings are reported with the algorithm parameters.

Search directions.

To explore different regions of the lower-level Pareto set, the algorithm cycles through a set of search-direction vectors $\mathbf{w} = (w_d, w_g, w_t)$, where w_d , w_g , and w_t represent the current emphasis on travel distance, active-workload inequality, and maximum route completion time, respectively. These weights guide current-state acceptance and selected search mechanisms, but they do not replace Pareto dominance in archive management. Consequently, the archive remains a nondominated representation of the lower-level objective space rather than a collection of solutions optimized for a single scalarized objective.

Normalized candidate change.

For candidate s' and current solution s , define $\Delta Z_m = Z_m(s') - Z_m(s)$. Because Z_2 , Z_3 , and Z_4 have different numerical scales, their changes are normalized using archive-based ranges. For $m \in \{2, 3, 4\}$,

$$R_m = \max \left\{ 10^{-6}, \max_{q \in \mathcal{A}} Z_m(q) - \min_{q \in \mathcal{A}} Z_m(q) \right\}. \quad (37)$$

The search-direction-dependent candidate change is then

$$\Delta(s', s \mid \mathbf{w}) = \beta_v \Delta Z_1 + \sigma \left(w_d \frac{\Delta Z_2}{R_2} + w_g \frac{\Delta Z_3}{R_3} + w_t \frac{\Delta Z_4}{R_4} \right), \quad (38)$$

where β_v controls the penalty associated with a fleet-size increase and σ scales the lower-level contribution.

The fleet-change term ensures that solutions using additional vehicles are strongly discouraged, while the weighted lower-level terms allow the search to emphasize different regions of the trade-off space. The archive-based ranges in Eq. (37) are used only for simulated-annealing acceptance and are distinct from the normalization bounds used for hypervolume evaluation in the computational experiments.

Candidate acceptance.

A feasible candidate that improves the current solution under the fleet-first relation is accepted directly. In particular, a candidate using fewer vehicles is always preferred. At an equal fleet size, a candidate that Pareto dominates the current solution with respect to (Z_2, Z_3, Z_4) is also accepted directly. Otherwise, the candidate may be accepted according to a simulated-annealing rule based on Eq. (38). Let T denote the current search temperature. The acceptance probability is

$$P_{\text{acc}}(s', s) = \begin{cases} 1, & \Delta(s', s \mid \mathbf{w}) \leq 0, \\ \exp\left(-\frac{\Delta(s', s \mid \mathbf{w})}{T}\right), & \Delta(s', s \mid \mathbf{w}) > 0. \end{cases} \quad (39)$$

Thus, candidates that are unfavorable under the current search direction may occasionally become the new current solution, particularly at higher temperatures. This mechanism helps the algorithm escape local regions of the solution space without weakening the fleet-first structure of the reported archive.

Importantly, archive insertion and current-state acceptance are evaluated separately. A nondominated same-fleet candidate can be inserted into the archive even when it does not become the current solution. Conversely, a candidate that is unsuitable for archive retention, including a temporary higher-fleet solution, may still be accepted as the current state when the acceptance rule permits it. The former mechanism preserves the best fleet-prioritized trade-off set discovered so far, whereas the latter controls the trajectory of the search.

After each feasible candidate evaluation, the selected destroy and repair operators receive rewards according to the candidate’s archive status and acceptance outcome. At segment boundaries, these accumulated scores are used to update operator-selection probabilities, and the search moves to the next lower-level search direction. If the archive fails to improve for a prescribed number of segments, stagnation-based perturbation is triggered. The corresponding parameter values are reported in Section ?? and Appendix A.

4.3 Initial solution generation

The initialization procedure generates a set of structurally diverse feasible solutions from which the initial current solution and Pareto archive are constructed. Several deterministic customer-ordering rules are combined with randomized priority sequences to provide different spatial, temporal, and capacity-oriented starting points.

For each customer sequence, routes are constructed through sequential insertion. A customer is inserted into an existing route whenever a feasible position is available. If direct insertion would violate battery feasibility, charging-station-assisted insertions are considered. A new route is opened only when the customer cannot be feasibly inserted into any existing route. All generated solutions are fully evaluated with respect to routing, time-window, payload, battery, and charging constraints before archive initialization. Detailed pseudocode and the associated construction settings are provided in Appendix B.1.

4.4 Partial-recharging evaluation

Charging decisions are evaluated jointly with the route sequence rather than by assuming full recharging at every station visit to avoid unnecessary charging time. The evaluation consists of two passes. First, a backward pass computes the minimum battery level $E_{i,\text{req}}^k$ that vehicle k must have when departing from each route position i to complete the remaining route suffix, accounting for subsequent planned charging opportunities. If this requirement exceeds the battery capacity at any route position, that route is declared infeasible.

Second, a forward pass propagates arrival times and battery levels along the route. Let E_i^k denote the battery level of vehicle k upon arrival at charging station i . The required charging amount is $c_i^k = \max\{0, E_{i,\text{req}}^k - E_i^k\}$. The charging decision is feasible only if $0 \leq c_i^k \leq B_{\text{max}} - E_i^k$. Its corresponding duration is

$$\tau_i^k = \frac{c_i^k}{g_s(i)}. \quad (40)$$

The charging duration is incorporated into the forward schedule, and the station availability constraints are subsequently checked. Thus, a vehicle receives only the energy required by the remaining route rather than being fully recharged by default.

For a station-assisted insertion, the algorithm additionally checks whether the resulting detour and station-stop time can be absorbed by waiting at a downstream customer. Consider inserting station i between predecessor p and customer j on the route of vehicle k . Let Θ_j^k denote the arrival time at customer j under the direct connection $p \rightarrow j$. The available waiting slack is $\phi_j^k = \max\{0, e_j - \Theta_j^k\}$.

Let ω_i^k denote any waiting required before charging can begin at station i . The additional time caused by the station visit is

$$\Delta T_{pij}^{k,\text{station}} = t_{pi} + \omega_i^k + \tau_i^k + t_{ij} - t_{pj}. \quad (41)$$

When the station is immediately available, $\omega_i^k = 0$. If $0 \leq \Delta T_{pij}^{k,\text{station}} \leq \phi_j^k$, the station visit does not delay the start of service at customer j , because the additional detour and station-stop time replace waiting that would otherwise occur before e_j . This condition is used only as a local screening rule. Complete route feasibility is subsequently verified through full temporal, payload, battery, and charging propagation.

4.5 Destroy operators

The destroy phase removes selected customers or route segments from the current solution to create opportunities for route restructuring. The operator set combines targeted removal mechanisms with random diversification. Targeted operators address energy risk, route consolidation, temporal restrictions, and fleet reduction, whereas random removal provides a less directed mechanism for escaping repeatedly explored solution structures. A summary of the destroy operators and their detailed pseudocodes are provided in [Appendix B.2](#).

Energy-critical removal.

Algorithm 5 (detailed in Appendix B.2) defines the energy-critical removal operator. The operator targets customers associated with elevated downstream energy risk. A customer insertion may be feasible at its immediate position while still leaving insufficient battery for the remaining part of the route. Therefore, for each customer c on route r , the operator evaluates the battery state after visiting c relative to the energy required for subsequent travel and eventual return to the depot. Customers with comparatively high energy-risk scores are prioritized for removal. Once removed, these customers are returned to the unserved set U , allowing the repair phase to reconsider both their route positions and the associated charging-station decisions rather than merely correcting the locally observed energy deficiency.

Route-merging removal.

Algorithm 6 (detailed in Appendix B.2) describes the route-merging removal operator. Its purpose is to create opportunities for route consolidation by identifying routes whose customer sets appear potentially compatible in terms of demand, geographical proximity, and time-window characteristics. Customers belonging to selected routes are removed simultaneously and returned to the unserved set U . Importantly, the operator does not directly concatenate the selected routes. Instead, their customers are reconstructed by the subsequent repair phase, which can redistribute them across the remaining routes. This avoids committing prematurely to a specific merge order and provides greater flexibility for reducing the number of utilized routes.

Time-slack removal.

Algorithm 7 (detailed in Appendix B.2) presents the time-slack removal operator. This operator targets customers whose current positions strongly restrict the temporal flexibility of their routes. A customer with a tight time window or an unfavorable position in the service sequence can consume a substantial amount of downstream scheduling slack and thereby limit later customer or charging-station insertions. The operator therefore computes a removal score reflecting schedule scarcity, the number of downstream visits affected, potential time savings from removal, and waiting-time patterns. Customers with high scores are preferentially removed so that the repair phase can reconstruct routes with greater temporal flexibility and, potentially, more efficient charging-station placement.

Fleet-reduction removal.

Algorithm 8 (detailed in Appendix B.2) implements the fleet-reduction removal operator. Simply removing the route containing the fewest customers often fails to reduce fleet size because the remaining routes may not possess sufficient payload capacity, temporal slack, or energy flexibility to absorb all displaced customers. The operator therefore removes the customers of a selected route together with additional customers that create insertion bottlenecks in the remaining solution. By enlarging the removal set, the operator creates additional redistribution flexibility and increases the likelihood that all removed customers can be reinserted without reopening the eliminated route.

Random removal.

Algorithm 9 (detailed in Appendix B.2) implements the random removal operator. Unlike the targeted removal operators, random removal selects customers uniformly without considering energy consumption, temporal feasibility, route structure, or objective-specific scores. This mechanism introduces diversification by preventing the search from repeatedly modifying the same high-priority customers or route structures. The selected customers are removed to form a partial solution, thereby creating opportunities for the repair phase to explore alternative customer assignments and route configurations that may be difficult to reach through more directed destroy operators.

4.6 Repair operators

After the destroy phase, the repair phase reconstructs a complete solution by reinserting all customers in the unserved set U . The repair operators differ in the order in which customers are selected, the way feasible insertion positions are ranked, and the extent to which immediate insertion quality is traded against future feasibility. Direct customer insertion and charging-station-assisted insertion are evaluated within the same reconstruction process rather than as independent post-processing steps. A summary of the repair operators and their detailed pseudocodes are provided in Appendix B.3.

Beam-search repair.

Algorithm 10 (detailed in Appendix B.3) defines the beam-search repair operator. A purely greedy reconstruction can select an insertion sequence that appears attractive initially but later forces long charging detours or the creation of an additional route. To reduce this myopic behavior, the operator retains a restricted set of promising partial constructions rather than committing immediately to a single one.

The procedure uses two levels of search. At the customer-sequence level, candidate partial routes are expanded by adding unserved customers. At the charging level, the resulting route extensions are evaluated together with the charging-station placements required to maintain energy feasibility. Only the most promising partial states are retained after each expansion. The resulting route-building procedure is repeatedly applied until all customers in U have been assigned.

Competitive-station insertion.

Algorithm 11 (detailed in Appendix B.3) presents the competitive-station insertion operator. Evaluating direct insertion first and considering a charging station only when the direct alternative fails can overlook a station-assisted insertion with a better overall route impact. This operator therefore evaluates direct and charging-station-assisted alternatives concurrently.

For each selected customer, feasible direct insertion positions are compared with positions in which a charging-station visit is inserted jointly with the customer. The best feasible alternative according to the applicable insertion criterion is selected. If no feasible insertion into the existing routes can be identified, the operator calls the

beam-search route-building procedure of Algorithm 10 to construct a new feasible route.

Adaptive insertion.

Algorithm 12 (detailed in Appendix B.3) describes the adaptive insertion operator. Rather than using one fixed insertion criterion throughout the search, this operator selects its insertion measure according to the current search state and lower-level search direction. The insertion criterion may therefore emphasize route packing, travel distance, active-workload balance, or maximum route completion time. For the selected customer, the operator first searches for a feasible position according to the active insertion measure. When no satisfactory direct position is available, charging-station-assisted insertion is considered. A new route is created only when the customer cannot be accommodated by the current routes. In that case, the operator invokes the beam-search route-building procedure defined in Algorithm 10.

Regret- K repair.

Algorithm 13 (detailed in Appendix B.3) defines the regret- K repair operator. This operator addresses the risk that a customer with only a small number of good insertion opportunities may become difficult or infeasible to insert if those positions are occupied by other customers first.

For each unserved customer, the procedure evaluates its best K insertion alternatives and computes a regret value from the deterioration between the best insertion and the remaining alternatives. Customers with larger regret values are prioritized because delaying their insertion is expected to cause a larger loss of future flexibility. After the customer has been selected, candidate insertion positions may additionally favor well-utilized existing routes to discourage unnecessary route creation. This packing adjustment affects the choice of insertion position, whereas the customer-priority calculation remains based on the underlying insertion costs. If no feasible existing-route insertion is available, the beam-search route-building procedure of Algorithm 10 is used as the fallback mechanism.

4.7 Local search

The destroy–repair phase may leave several residual inefficiencies. In addition, opportunities to reduce the number of utilized vehicles may require coordinated modifications across multiple routes and may therefore remain undetected by the repair operators. As such, the local-search procedure (Algorithm 2 detailed in Appendix C) addresses these issues through three complementary stages:

1. routing and workload refinement through variable neighborhood descent;
2. charging-station visit optimization; and
3. route consolidation and fleet reduction.

These stages are applied iteratively until the prescribed stopping condition is met. The routing stage provides frequent low-disruption improvements, while charging optimization and fleet reduction are activated when the current solution structure justifies their higher computational cost.

Algorithm 2: Local-search procedure

Input: Feasible solution s **Output:** Locally improved solution s

```
1 while  $\neg$  LocalSearchStoppingCriterionMet() do
2    $s \leftarrow$  RoutingVND( $s$ );
3   if ChargingOptimizationTriggered( $s$ ) then
4      $s \leftarrow$  OptimizeChargingVisits( $s$ );
5   if FleetReductionTriggered( $s$ ) then
6      $s \leftarrow$  ReduceFleetSize( $s$ );
7   UpdateLocalSearchState( $s$ );
8 return  $s$ ;
```

Routing and workload refinement

The first stage applies variable neighborhood descent (VND) to improve customer sequencing and assignment. The neighborhoods include both intra-route and inter-route moves, allowing the search to reduce travel distance, redistribute active workload, and improve route completion times without changing the number of utilized vehicles. Specifically, the neighborhoods are examined according to increasing structural disruption. Under the first-improvement strategy, the search accepts the first feasible improving move and restarts from the first neighborhood. This restart mechanism emphasizes inexpensive local refinements before more disruptive moves are explored. Table 2 summarizes different neighborhood structures used in this stage.

Charging and fleet-structure refinement

The remaining local-search stages refine charging-station visits and route assignments. Charging visits introduced during construction or repair may become unnecessary or inefficient after the customer sequence changes. The algorithm therefore considers removing, relocating, replacing, or inserting station visits while keeping the customer assignment fixed. Each modification is evaluated using the partial-recharging procedure in Section 4.4; charging amounts and route schedules are recomputed, and only feasible accepted improvements or Pareto trade-offs are retained.

The fleet-reduction stage attempts to improve the primary objective Z_1 by eliminating routes and reassigning their customers to the remaining fleet. The procedure first attempts direct route emptying and smallest-route elimination. When these operations fail, it considers more coordinated moves, including ejection chains, segment exchanges, multi-route consolidation, and three-to-two route merging. Any solution using fewer vehicles is fully evaluated before the Pareto archive is updated. A move that preserves the fleet size may also be retained if it improves the lower-level objectives or facilitates a later fleet reduction.

Table 2 Neighborhood structures used in the routing VND.

Neighborhood	Move	Scope
Relocate	Moves one customer to another position.	Intra-route or inter-route.
Swap	Exchanges the positions of two customers.	Intra-route or inter-route.
Inter-route Or-opt	Transfers a short consecutive customer segment between routes.	Inter-route.
Intra-route 2-opt	Reverses a subsequence within one route.	Intra-route.
Inter-route 2-opt	Exchanges the terminal portions of two routes.	Inter-route.
Intra-route Or-opt	Relocates a consecutive customer segment within the same route.	Intra-route.
Cross-exchange	Exchanges customer segments between two routes.	Inter-route.
Reversed Or-opt	Transfers a customer segment to another route while reversing its order.	Inter-route.

5 Computational Experiments

Having specified the FULL framework and its search mechanisms, we now assess whether explicit equity guidance can yield practically meaningful active-workload improvements without compromising fleet-size or travel-distance performance. The computational study addresses four questions. First, we assess whether the proposed equity-aware framework remains competitive with published benchmark solutions in terms of fleet size and travel distance (**RQ1**). Second, we examine the system-level effect of explicit equity guidance by comparing FULL with a controlled No-EQ ablation at a common fleet level (**RQ2**). Third, we quantify the active-workload equity attainable when management permits a limited increase in travel distance (**RQ3**). Finally, we examine whether lower Gini values correspond to changes in route-level workload concentration and dispersion (**RQ4**). The section concludes with two illustrative case studies.

The experiments use the 56 100-customer E-VRPTW instances of [Schneider et al. \(2014\)](#) under the partial-recharging setting of [Keskin and Catay \(2016\)](#). The benchmark contains the six standard classes C1, C2, R1, R2, RC1, and RC2, with 21 charging-station locations in each instance. All experiments were executed using one CPU thread on an Intel Core i5-12500H machine with 64 GB RAM under Linux Mint. The C++ implementation was compiled with GCC using the `-O3` optimization flag.

Each algorithmic variant was run 10 times per instance using common random seeds $\{1, \dots, 10\}$ and a fixed budget of 25,000 iterations per run. Thus, the repeated

runs characterize the stochastic behavior of the search within each benchmark instance, whereas the benchmark instance is the statistical unit for paired inference across variants. Table 3 summarizes the experimental protocol and search budget.

Table 3 Experimental protocol and iteration budget.

Setting	Value
Number of instances	56
Runs per instance and variant	10
Random seeds	[1, 10]
Iterations per run	25,000
CPU threads	1
Total runs per variant	560

Common-fleet comparison protocol

Because workload inequality is mechanically affected by the number of active routes, lower-level comparisons are conducted only at a fleet size that both variants attain with sufficient reliability. A run is considered valid for a particular analysis if its final archive contains at least one solution satisfying all fleet and budget conditions required by that analysis. Consequently, run validity is comparison-specific. Runs that do not satisfy the relevant conditions are treated as unavailable for that comparison rather than assigned an artificial objective penalty.

For instance i , variant v , and run r , let $\mathcal{F}_{i,v}^{(5)}$ denote the set of fleet sizes attained by variant v in at least five of its ten runs: $\mathcal{F}_{i,v}^{(5)} = \{k : \# \{r : \exists s \in \mathcal{A}_{ivr}, Z_1(s) = k\} \geq 5\}$. The common fleet target is then $Z_{1,i}^{\text{common}} = \min(\mathcal{F}_{i,\text{FULL}}^{(5)} \cap \mathcal{F}_{i,\text{NoEQ}}^{(5)})$. This definition selects the smallest fleet size that is attained by both variants in at least half of their runs. Therefore, differences in Z_2 , Z_3 , or Z_4 observed in the subsequent common-fleet analyses cannot be attributed to either variant using more vehicles.

The criterion is satisfied for 55 of the 56 benchmark instances. Instance c103_21 is excluded from the common-fleet lower-level comparisons because no fleet level is attained by both variants in at least five runs. Specifically, No-EQ reaches $Z_1 = 10$ in 6 of 10 runs whereas FULL reaches that level in only 3 runs; conversely, FULL reaches $Z_1 = 11$ in 7 runs whereas No-EQ reaches that level in only 4. Hence, 55 instances are retained for **RQ2**, **RQ3**, and **RQ4**, while the benchmark competitiveness analysis for **RQ1** continues to use all 56 instances.

Table 4 reports the success rate of each variant at the corresponding common fleet target. For a given instance, the success rate is the proportion of the ten runs whose final archive contains at least one solution with $Z_1 = Z_{1,i}^{\text{common}}$. By construction, both variants attain at least a 50% success rate on every instance retained in the common-fleet analysis.

Table 4 Fleet-target success rates at the common fleet level $Z_{1,i}^{\text{common}}$.

Series	N_{inst}	FULL SR (%)	No-EQ SR (%)
C1	8	100.00	98.75
C2	8	100.00	100.00
R1	12	86.67	89.17
R2	11	100.00	100.00
RC1	8	83.75	86.25
RC2	8	95.00	95.00
Unweighted mean	–	94.24	94.86

Statistical analysis

Repeated-run results are first summarized within each benchmark instance, and paired inference is then conducted across matched instances. Unless otherwise stated, instance-level summaries are based on the valid runs for the corresponding comparison. Paired tests require that both variants meet the analysis-specific validity condition for the instance.

Wilcoxon signed-rank tests are used for paired instance-level comparisons. The reported N_{paired} is the number of matched instances and n_{nonzero} is the effective sample size after zero paired differences are removed. Rank-biserial correlation is reported as the effect size, with the sign oriented so that positive values favor FULL when appropriate. Confidence intervals are obtained from 10,000 bootstrap resamples. For the five endpoints jointly reported in the main RQ2 ablation analysis, multiplicity is controlled using Holm adjustment. For the four positive distance-budget tests in RQ3, multiplicity is controlled using a Bonferroni adjustment. The 0% distance-budget point in RQ3 is descriptive only and is not part of that multiplicity family.

5.1 Fleet-size and routing competitiveness against benchmark solutions (RQ1)

We first assess whether introducing explicit equity guidance compromises the ability of the search to attain competitive fleet sizes and travel distances. Using the calibrated configuration described above, FULL and No-EQ are compared with the published best-known solutions (BKS) of Keskin and Catay (2016).

FULL and No-EQ match the published best-known fleet size on 43 and 44 of the 56 instances, respectively. Their average fleet-size gaps are +0.20 and +0.18 vehicles. The corresponding average travel-distance gaps are +0.46% for FULL and +0.13% for No-EQ.

The distance gaps should be interpreted descriptively rather than as strict like-for-like measures of routing efficiency because the fleet size attained by a variant does not coincide with the BKS fleet size on every instance. In particular, a negative distance gap obtained with a larger fleet should not be interpreted as a routing improvement at the same fleet level. For this reason, all subsequent lower-level comparisons between FULL and No-EQ use the common-fleet protocol defined earlier in this section.

Overall, the benchmark results indicate that introducing explicit active-workload equity guidance does not materially compromise fleet-size competitiveness. FULL remains close to both No-EQ and the published fleet-size references while maintaining relatively small descriptive distance deviations. This establishes a suitable baseline for the common-fleet equity analyses that follow.

Table 5 Descriptive comparison of attained fleet size and travel distance against published best-known solutions. Travel-distance gaps are not necessarily computed at identical fleet sizes on every instance.

Series	Fleet Size (Z_1)		Travel Distance (Z_2)	
	FULL	No-EQ	FULL	No-EQ
C1	+1.03%	+0.00%	+0.59%	+0.49%
C2	+0.00%	+0.00%	+0.00%	+0.00%
R1	+1.91%	+1.30%	+0.50%	+0.01%
R2	+13.64%	+13.64%	-2.15%	-2.12%
RC1	+0.92%	+1.92%	+4.95%	+4.34%
RC2	+8.33%	+8.33%	-1.88%	-2.44%
Avg.	+2.53%	+2.28%	+0.46%	+0.12%

5.2 Value of explicit equity guidance (RQ2)

RQ2 examines whether actively including workload inequality in the search changes the quality and composition of the decision set at a common fleet size. We answer this in three stages. First, we define the controlled No-EQ ablation. Second, we compare one distance-focused representative per run. Third, we compare the aggregate Pareto-set coverage obtained by the two variants in common ex-post evaluation spaces.

5.2.1 Controlled No-EQ ablation

Both variants compute and report the complete objective vector (Z_1, Z_2, Z_3, Z_4) , but they differ in which lower-level objectives actively govern the search. Conditional on fleet size, define the active lower-level objective index set

$$\mathcal{J}_v = \begin{cases} \{2, 3, 4\}, & v = \text{FULL}, \\ \{2, 4\}, & v = \text{NoEQ}. \end{cases} \quad (42)$$

For variant v , a same-fleet solution s' dominates s if

$$Z_j(s') \leq Z_j(s) \quad \forall j \in \mathcal{J}_v, \quad Z_{j^*}(s') < Z_{j^*}(s) \text{ for at least one } j^* \in \mathcal{J}_v. \quad (43)$$

Thus, FULL maintains its active search archive on (Z_2, Z_3, Z_4) , whereas No-EQ maintains its archive on (Z_2, Z_4) . No-EQ still computes Z_3 for ex-post analysis, but Gini does not protect a solution from removal in the active No-EQ archive. Consequently, a solution with a substantially lower workload inequality but slightly worse distance or completion time may remain relevant to FULL while being discarded by No-EQ.

The same distinction is applied to the other search-control channels. Relative to FULL, No-EQ disables the Gini contribution in simulated-annealing acceptance, equity-oriented objective-weight transitions, equity-aware operator rewards, and equity-aware perturbation. All remaining algorithmic components and the FULL-calibrated parameter values are held fixed. The ablation therefore measures the combined system-level effect of explicit equity guidance; it does not isolate the marginal contribution of each individual equity-aware mechanism. Table 6 summarizes the controlled differences.

Table 6 Search-active differences and common elements of FULL and No-EQ.

Component	FULL	No-EQ
Computed and reported objectives	(Z_1, Z_2, Z_3, Z_4)	(Z_1, Z_2, Z_3, Z_4)
Search archive at fixed fleet size	(Z_2, Z_3, Z_4)	(Z_2, Z_4)
Gini in SA acceptance	Active	Disabled
Equity-oriented weight channels	Active	Disabled
Equity-aware operator rewards	Active	Disabled
Equity-aware perturbation	Active	Disabled
Ex-post Gini calculation	Retained	Retained
Common parameters	FULL-calibrated	Same inherited values
Independent recalibration	Not applicable	No

5.2.2 Distance-focused representative solutions

We first examine whether explicit equity guidance changes a practically relevant solution near the distance-minimizing region. For instance i , variant v , and run r , let $s_{ivr}^{\text{TD}} \in \arg \min\{Z_2(s) : s \in \mathcal{A}_{ivr}, Z_1(s) = Z_{1,i}^{\text{common}}\}$ denote the distance-focused representative, with ties broken lexicographically by Z_3 and then Z_4 . This tie-breaking rule is used only for ex-post representative selection and does not affect the No-EQ search.

Table 7 reports the median objective values of these representatives. Because fleet size is fixed by construction, any difference in travel distance or workload inequality reflects a lower-level routing trade-off rather than additional fleet capacity.

For each matched instance, define the relative travel-distance difference as $\Delta_{i,Z_2}^{\%} = 100(Z_{2,i}^{\text{FULL}} - Z_{2,i}^{\text{NoEQ}})/Z_{2,i}^{\text{NoEQ}}$. The median instance-level difference is approximately 0.4%. Thus, under the common-iteration and shared-configuration protocol, FULL incurs a small distance premium in the distance-focused region while attaining

Series	Distance (Z_2)		Workload Gini (Z_3)		Max. completion time (Z_4)	
	FULL	No-EQ	FULL	No-EQ	FULL	No-EQ
C1	1011.77	1002.79	0.0755	0.0727	1215.32	1208.68
C2	629.95	629.95	0.1553	0.1553	3360.78	3360.78
R1	1228.46	1221.51	0.0923	0.0961	228.40	228.05
R2	895.00	895.33	0.0407	0.0752	951.23	953.26
RC1	1403.62	1398.91	0.0959	0.1066	238.02	236.98
RC2	1106.40	1099.66	0.0704	0.0807	919.14	925.56

Table 7 Median objective values of distance-focused solutions at the common fleet level.

lower workload inequality. No systematic shift in maximum route completion time is detected.

Figure 1 shows the paired instance-level Gini differences, with negative values favoring FULL. The figure illustrates that the equity effect is not driven by a small number of benchmark classes or isolated instances. The corresponding paired statistical results are reported jointly with the Pareto-coverage endpoints in Table 9 after the full RQ2 evaluation space has been defined.

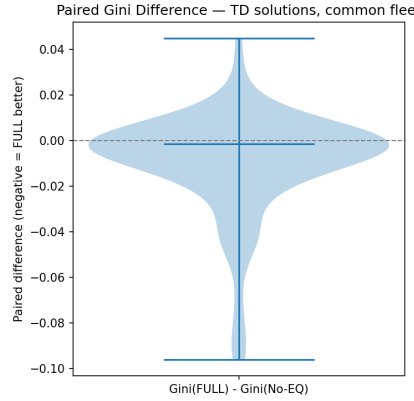


Figure 1 Paired Gini differences ($\Delta Z_3 = Z_3^{\text{FULL}} - Z_3^{\text{NoEQ}}$) for distance-focused solutions at the common fleet level.

5.2.3 Equity-aware Pareto-set coverage

The representative comparison above examines only one solution per run and therefore does not show whether the equity advantage extends across the attainable decision set. To evaluate aggregate trade-off coverage, we compare the pooled final archives of FULL and No-EQ using common ex-post hypervolume spaces. These evaluation indicators are denoted by HV^{eval} and are distinct from the internal HV^{online} and HV^{cal} quantities defined in Appendix A.1.

For each instance i and variant v , let $\mathcal{R}_{iv}$ be the union of the 10 final run archives restricted to the common fleet level $Z_{1,i}^{\text{common}}$. Since nondominance depends on the objectives included in the evaluation, separate nondominated sets are constructed for the two common evaluation spaces:

$$\mathcal{P}_{iv}^{2D} = \text{ND}_{(Z_2, Z_3)}(\mathcal{R}_{iv}), \quad (44)$$

$$\mathcal{P}_{iv}^{3D} = \text{ND}_{(Z_2, Z_3, Z_4)}(\mathcal{R}_{iv}). \quad (45)$$

For evaluation dimension $d \in \{2D, 3D\}$, define the pooled set $\mathcal{P}_i^d = \mathcal{P}_{i,\text{FULL}}^d \cup \mathcal{P}_{i,\text{NoEQ}}^d$. Let $\mathcal{J}_{2D} = \{2, 3\}$, $\mathcal{J}_{3D} = \{2, 3, 4\}$. For each objective $j \in \mathcal{J}_d$, common lower and upper bounds are obtained from the pooled set:

$$L_{ij}^{\text{eval},d} = \min_{s \in \mathcal{P}_i^d} Z_j(s), \quad U_{ij}^{\text{eval},d} = \max_{s \in \mathcal{P}_i^d} Z_j(s). \quad (46)$$

Each objective value is normalized as

$$\hat{Z}_{ij}^d(s) = \begin{cases} \frac{Z_j(s) - L_{ij}^{\text{eval},d}}{U_{ij}^{\text{eval},d} - L_{ij}^{\text{eval},d}}, & U_{ij}^{\text{eval},d} > L_{ij}^{\text{eval},d}, \\ 0, & U_{ij}^{\text{eval},d} = L_{ij}^{\text{eval},d}. \end{cases} \quad (47)$$

Let $\hat{\mathcal{P}}_{iv}^{2D}$ and $\hat{\mathcal{P}}_{iv}^{3D}$ denote the corresponding normalized sets. The evaluation indicators are

$$HV_{iv}^{\text{eval},2D} = HV_2(\hat{\mathcal{P}}_{iv}^{2D}; (1.05, 1.05)), \quad (48)$$

$$HV_{iv}^{\text{eval},3D} = HV_3(\hat{\mathcal{P}}_{iv}^{3D}; (1.05, 1.05, 1.05)). \quad (49)$$

Because both variants use the same pooled bounds within each instance, their ex-post evaluation hypervolumes are directly comparable for that instance even though the active search archives differ. In particular, No-EQ searches in (Z_2, Z_4) at a fixed fleet size but is evaluated ex post in the same (Z_2, Z_3) and (Z_2, Z_3, Z_4) spaces as FULL. The pooled-run hypervolumes quantify the aggregate decision-set coverage attained across the 10-run experiment. They should therefore be interpreted as measures of attainable coverage across repeated runs rather than as the expected hypervolume of a single run.

Table 8 shows that FULL obtains a larger mean instance-level evaluation hypervolume in every benchmark class in both evaluation spaces. Figure 2 illustrates the corresponding (Z_2, Z_3) projections for representative clustered, random, and mixed instances. Part of the hypervolume difference follows directly from the archive design. No-EQ does not retain a solution solely because it has a favorable Gini value when that solution is dominated in its active (Z_2, Z_4) space, whereas FULL explicitly protects alternatives that are nondominated in the equity-inclusive space. Hence, the larger equity-inclusive hypervolumes should be interpreted as the system-level effect of equity-aware search guidance and archive retention on the attainable decision set.

Table 8 Mean instance-level evaluation hypervolumes within each benchmark class under the common pooled normalization.

Class	HV_{2D}^{eval}		HV_{3D}^{eval}	
	FULL	No-EQ	FULL	No-EQ
C1	0.8513	0.5642	0.9073	0.6621
C2	0.9391	0.5082	0.9681	0.6126
R1	0.8324	0.5740	0.7585	0.5386
R2	0.9535	0.6668	0.9828	0.8533
RC1	0.7759	0.5441	0.7354	0.5084
RC2	0.7996	0.5360	0.9259	0.8011

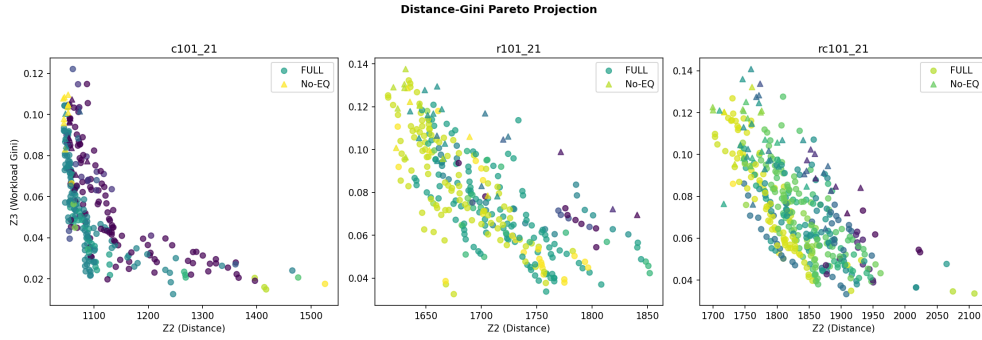


Figure 2 Representative (Z_2, Z_3) projections of the pooled nondominated sets at the common fleet level for C101_21, R101_21, and RC101_21. Circles denote FULL and triangles denote No-EQ.

They are not criterion-independent evidence that FULL is universally superior, nor do they imply that FULL is better when distance or maximum completion time is considered in isolation.

5.2.4 Paired statistical synthesis

Table 9 summarizes the paired instance-level comparisons for the three representative-solution objectives and the two evaluation-hypervolume endpoints. N_{paired} denotes the number of matched instances, n_{nonzero} the effective Wilcoxon sample size after zero differences are removed, and r_{rb} the rank-biserial correlation. F/T/N counts instances where FULL is better, tied, or worse than No-EQ. Confidence intervals are based on 10,000 bootstrap resamples.

The results show statistically detectable differences in total distance (Z_2), workload inequality (Z_3), HV_{2D}^{eval} , and HV_{3D}^{eval} , whereas no systematic difference is detected for maximum route completion time (Z_4). A Holm adjustment across the five reported endpoints leaves these conclusions unchanged. Taken together, the representative and

decision-set-level results indicate that explicit equity guidance changes both the location and the coverage of the attainable lower-level trade-off set at a fixed fleet size.

Table 9 Paired statistical comparison between FULL and No-EQ at the common fleet level.

Metric	N_{paired}	n_{nonzero}	Median Diff	95% CI (Bootstrap)	p-value	$r_{rb}^\dagger$	F / T / N
TD_Z2	55	47	+3.7636	[0.2841, 4.8312]	5.13e-04	-0.5656	11 / 8 / 36
TD_Z3	55	47	-0.0038	$[-0.0103, -4.0 \times 10^{-5}]$	3.43e-03	+0.4823	33 / 8 / 14
TD_Z4	55	39	0.0000	[0.0000, 0.6572]	4.60e-01	-0.1385	16 / 16 / 23
HV_{2D}^{eval}	55	55	+0.2669	[0.2282, 0.3208]	1.92e-10	+0.9870	53 / 0 / 2
HV_{3D}^{eval}	55	55	+0.2005	[0.1280, 0.2561]	2.39e-10	+0.9818	54 / 0 / 1

5.3 Budgeted distance–equity trade-offs (RQ3)

The preceding analyses compare either one representative point or the unconstrained pooled decision set. RQ3 instead considers a more operational question: how much workload equity can be obtained when management imposes a limit on additional routing distance and on maximum route completion time? For each instance i , variant v , and run r , define the best attainable Gini under the corresponding budgets as

$$g_{ivr}(\eta_D, \eta_H) = \min\{Z_3(s) : Z_1(s) = Z_{1,i}^{\text{common}}, \\ Z_2(s) \leq (1 + \eta_D)D_i^{\text{ref}}, Z_4(s) \leq (1 + \eta_H)H_i^{\text{ref}}\}, \quad (50)$$

where $\eta_D \in \{0, 1\%, 3\%, 5\%, 10\%\}$, $\eta_H = 5\%$. Here, η_D is the allowable travel-distance premium and η_H is the allowable maximum-completion-time premium. The reference values D_i^{ref} and H_i^{ref} are the component-wise minimum pooled distance and completion time at the common fleet. They therefore act as ideal anchor coordinates and need not be attained simultaneously by the same archived solution.

Inferential tests are conducted for the four positive distance budgets 1%, 3%, 5%, and 10%. The 0% point is retained for descriptive reference only because its matched-sample requirement is not satisfied. Bonferroni correction is applied across the four positive-budget tests.

Figure 3 plots the resulting budgeted-equity curve, and Table 10 reports the paired statistical comparisons. Across matched instances, FULL attains lower median Gini values at all positive distance budgets. The difference becomes statistically significant from the 3% budget onward after Bonferroni correction ($p_{\text{Bonf}} \leq 5.08 \times 10^{-6}$, $r_{rb}^\dagger \geq 0.78$). At the 1% budget, FULL also has a lower median Gini ($p = 0.032$, $r_{rb}^\dagger = 0.47$), but this difference does not remain significant after multiplicity correction.

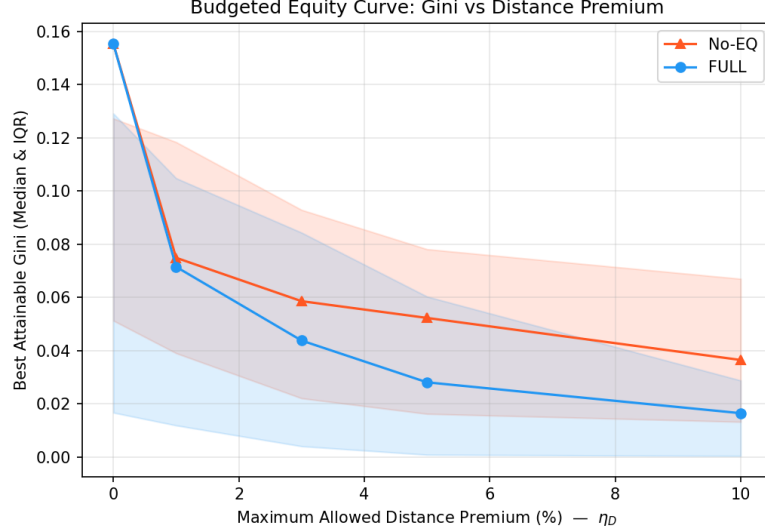


Figure 3 Best attainable median Gini as a function of allowable travel-distance premium (η_D).

Table 10 Paired statistical comparison of best attainable Gini under travel-distance budgets ($\eta_H = 5\%$).

Budget	η_D	N_{paired}	n_{nonzero}	FULL Median	No-EQ Median	Median Diff	p-value	p_{Bonf}	$r_{rb}^\dagger$
1%		27	27	0.0715	0.0749	-0.0074	3.15e-02	1.26e-01	+0.4735
3%		43	43	0.0438	0.0586	-0.0120	1.27e-06	5.08e-06	+0.7844
5%		49	49	0.0281	0.0523	-0.0147	5.62e-10	2.25e-09	+0.8955
10%		55	55	0.0165	0.0365	-0.0216	1.11e-10	4.44e-10	+1.0000

Operationally, these results identify a practically relevant region of the distance–equity trade-off. A 1% distance allowance already shows a tendency toward lower workload inequality under FULL, although the evidence is not strong enough to survive multiplicity correction. Once the allowable distance premium reaches 3%, the advantage of explicit equity guidance becomes both statistically strong and substantively consistent. Thus, for the benchmark instances, a modest relaxation of pure distance minimization can unlock substantially more balanced active-workload allocations.

The matched-instance set may expand as the distance budget increases because more runs satisfy the joint fleet, distance, and completion-time conditions. Accordingly, Figure 3 should be interpreted as a sequence of budget-specific attainment summaries rather than as repeated measurements on one fixed set of benchmark instances.

5.4 Route-level workload diagnostics (RQ4)

RQ4 is a diagnostic check on the substantive interpretation of the Gini objective rather than an additional optimization comparison. A lower relative inequality value does not imply that every route has a smaller workload, and the Gini coefficient does not by itself identify whether the change is driven by one extreme route or by a broad narrowing of the workload distribution. We therefore examine two complementary route-level measures.

First, the maximum-to-mean ratio measures the concentration of active workload on the most heavily loaded route:

$$\rho_{\max} = \frac{\max_{k \in K^+} A_k^{\text{act}}}{\bar{A}^{\text{act}}}. \quad (51)$$

Second, the coefficient of variation measures fleet-wide workload dispersion:

$$\text{CV} = \frac{1}{\bar{A}^{\text{act}}} \sqrt{\frac{1}{|K^+|} \sum_{k \in K^+} (A_k^{\text{act}} - \bar{A}^{\text{act}})^2}. \quad (52)$$

The two diagnostics capture different operational patterns. $\rho_{\max}$ is sensitive to an extreme-route concentration effect, whereas CV summarizes dispersion across the entire active fleet. Together they provide a route-level check on whether the lower Gini values observed under FULL are accompanied by a narrower workload distribution.

For each valid run, the feasible archive solution with minimum Z_3 is selected under the 3% distance budget and 5% completion-time budget. The two diagnostics are computed for that solution and summarized descriptively across valid runs within each benchmark class. These pooled run-level summaries are not used for instance-level inference.

As shown in Table 11, the lower Gini values obtained by FULL are generally accompanied by lower $\rho_{\max}$ and CV, with the clearest differences for C2, R1, and RC1. For RC2, FULL yields lower median Gini, median $\rho_{\max}$, and median CV, although the corresponding mean values are nearly identical across variants. The patterns for C1 and R2 are more modest or mixed. These diagnostics are therefore consistent with, but do not constitute an additional inferential proof of, broader route-level workload redistribution.

5.5 Illustrative case studies

The preceding analyses establish the aggregate effects of explicit equity guidance. The following two cases illustrate the route-level mechanisms behind these results. The first compares FULL and No-EQ under identical operational budgets, whereas the second shows how active-workload distribution changes along the FULL distance–equity Pareto front.

Table 11 Descriptive route-level workload-distribution metrics at the 3% distance budget ($\eta_H = 5\%$).

Series	Variant	Workload Gini				$\rho_{\max}$		CV	
		Min	Mean	Median	Max	Mean	Median	Mean	Median
C1	FULL	0.0062	0.0302	0.0292	0.0735	1.0780	1.0742	0.0585	0.0556
C1	No-EQ	0.0056	0.0333	0.0309	0.0813	1.0809	1.0703	0.0643	0.0594
C2	FULL	0.0009	0.0565	0.0439	0.1559	1.1128	1.0918	0.1090	0.0811
C2	No-EQ	0.0036	0.0926	0.0781	0.1559	1.1712	1.1448	0.1813	0.1517
R1	FULL	0.0079	0.0296	0.0297	0.0680	1.0675	1.0676	0.0546	0.0544
R1	No-EQ	0.0223	0.0389	0.0371	0.0726	1.0803	1.0781	0.0742	0.0709
R2	FULL	0.0001	0.0085	0.0061	0.0365	1.0182	1.0122	0.0164	0.0116
R2	No-EQ	0.0001	0.0096	0.0057	0.0528	1.0192	1.0130	0.0186	0.0110
RC1	FULL	0.0058	0.0399	0.0393	0.0750	1.0856	1.0881	0.0761	0.0764
RC1	No-EQ	0.0228	0.0492	0.0462	0.0877	1.0929	1.0933	0.0957	0.0875
RC2	FULL	0.0008	0.0148	0.0103	0.0548	1.0316	1.0216	0.0284	0.0198
RC2	No-EQ	0.0004	0.0144	0.0117	0.0563	1.0317	1.0261	0.0276	0.0222

FULL vs. No-EQ route-level comparison.

For each variant, we select the minimum- Z_3 solution from the nondominated union of the 10 run archives subject to $Z_1 = Z_{1,i}^{\text{common}}$, $Z_2 \leq 1.03D_i^{\text{ref}}$, $Z_4 \leq 1.05H_i^{\text{ref}}$. Ties in Z_3 are broken first by Z_2 and then by Z_4 . Applying the same fleet, distance, and completion-time conditions to both variants ensures that the comparison is not driven by different operational budgets. For route k , active workload is $A_k^{\text{act}} = T_k^{\text{travel}} + T_k^{\text{service}} + T_k^{\text{charging}}$, whereas total route completion time additionally includes waiting, $H_k = A_k^{\text{act}} + W_k$.

For instance R101_21, both variants use 18 routes. FULL reduces the active-workload Gini coefficient from 0.0947 to 0.0371, corresponding to a relative reduction of approximately 60.8%, while total travel distance increases only from 1660 to 1668, or 0.48%. By contrast, the route-completion distributions remain similar: mean H_k is 211.5 for FULL and 211.3 for No-EQ, with standard deviations of approximately 13 in both cases. FULL has a slightly larger maximum completion time (229 versus 226), but a slightly narrower completion-time range (41 versus 44).

Figure 4 therefore illustrates that the lower Gini obtained by FULL reflects a redistribution of travel, service, and charging workload across routes rather than a general equalization of route completion times. This supports the separate treatment of active-workload inequality Z_3 and maximum route completion time Z_4 .

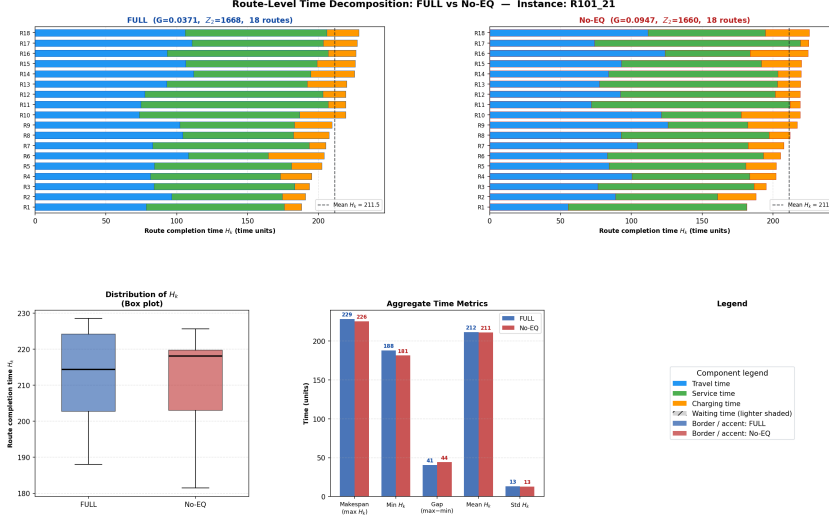


Figure 4 Route-level time decomposition for instance R101_21 under the common budget-constrained selection rule. Stacked bars report travel, service, charging, and, when present, waiting time for FULL and No-EQ. Routes are sorted independently by total completion time H_k within each variant, so adjacent bars across panels do not represent matched customer sets.

Within-FULL Pareto trade-off.

Figure 5 provides a complementary view of the distance–equity trade-off within the pooled FULL Pareto set for instance R104_21. Three representative solutions are highlighted: the minimum-distance solution, the elbow trade-off solution, and the minimum-Gini solution. The elbow is defined in the normalized (Z_3, Z_2) space as the archived point with maximum perpendicular distance from the line connecting the minimum-distance and minimum-Gini endpoints.

The minimum-distance solution has $(Z_2, Z_3) = (1100, 0.0800)$. Moving to the elbow yields $(Z_2, Z_3) = (1163, 0.0300)$, which reduces Gini by 62.5% at a distance premium of 5.73%. Relative to the total attainable reduction from 0.0800 to 0.0100, the elbow captures approximately 71.4% of the Gini improvement.

The minimum-Gini solution reaches $Z_3 = 0.0100$ at $Z_2 \approx 1335$, corresponding to an overall distance premium of approximately 21.4% relative to the minimum-distance solution. At this equity-oriented extreme, route active workloads lie within the narrow interval of approximately 216–227 time units around a mean of 222.2.

The route-level profiles therefore show a progressive narrowing of the active-workload distribution as the solution moves toward the equity-oriented end of the Pareto front. For this instance, the elbow captures most of the attainable equity improvement at substantially lower routing cost than the minimum-Gini extreme, consistent with the budgeted distance–equity results in Section 5.3.

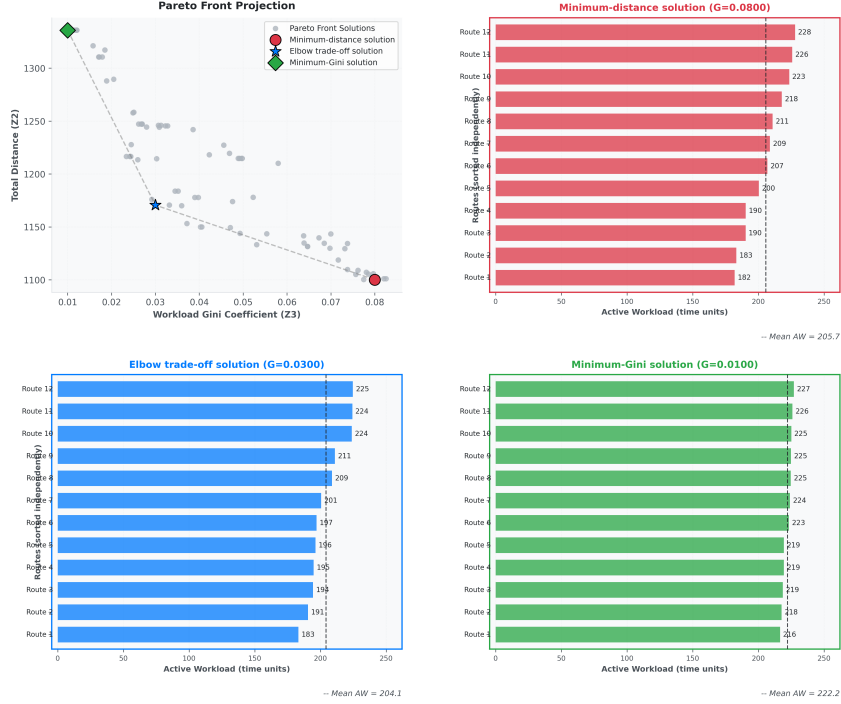


Figure 5 Within-FULL distance-equity trade-off for instance R104_21. The top-left panel shows the (Z_3, Z_2) projection of the archived nondominated solutions at the selected fleet level. The remaining panels report route-level active workload A_k^{act} for the minimum-distance, elbow trade-off, and minimum-Gini solutions. Routes are sorted independently by active workload within each solution, and the dashed vertical line denotes the corresponding mean.

6 Conclusion

This paper studies active-workload equity in the E-VRPTW-PR using a fleet-prioritized multi-objective Pareto ALNS. Across 56 benchmark instances and 10 runs per variant, explicit equity guidance generated lower-inequality solutions at common fleet levels. For distance-focused representatives, the median travel-distance premium of FULL was approximately 0.4%, while the budgeted analysis showed increasingly strong equity advantages from a 3% distance allowance onward. No systematic difference in maximum route completion time was detected under the adopted protocol.

The results indicate that explicit equity guidance changes the composition of the attainable decision set rather than simply replacing distance efficiency with workload balance. Its benefit is not uniform across all instance classes, but the overall pattern is consistent across distance-focused representatives, equity-inclusive Pareto coverage, budget-constrained comparisons, and route-level workload diagnostics. The

case studies further illustrate that improved active-workload equity does not necessarily imply more equal route completion times and that much of the attainable equity improvement may occur in an intermediate region of the distance–equity Pareto front.

From an operational perspective, three implications emerge. First, distance-efficient routes do not necessarily distribute travel, service, and charging duties evenly. Second, the 3%–5% distance-premium region appears particularly relevant for the benchmark instances, as substantial equity improvements can often be obtained without evidence of a systematic increase in maximum route completion time. Third, intermediate Pareto solutions may offer more attractive compromises than either the minimum-distance or minimum-Gini extremes.

The comparison should nevertheless be interpreted within the scope of the experimental design. No-EQ inherits the configuration calibrated for FULL, so the study evaluates the system-level effect of disabling equity-guided archive management, acceptance, search-direction transitions, operator rewards, and perturbation within a shared framework. It does not establish superiority over an independently calibrated equity-blind algorithm. Moreover, the adopted active-workload measure is an operational proxy rather than a direct measure of fatigue or compensable labor, and a lower Gini does not imply that every route workload decreases.

Future work may therefore consider alternative workload definitions, independently or jointly calibrated variants, component-wise ablations, and other operational settings in which workload equity interacts with charging, time windows, and routing efficiency.

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A Parameter tuning and settings

This appendix reports the complete parameter-tuning procedure and the configuration used throughout the computational experiments. We first describe the calibration criterion and the hypervolume quantities used during tuning, followed by the calibrated parameter domains and selected values. The internal settings that were fixed before calibration are then reported separately.

A.1 Calibration procedure

The `irace` package (López-Ibáñez et al. 2016) was used to select an empirical parameter configuration for FULL. The purpose of the calibration was to identify a single configuration for the subsequent computational study rather than to establish globally optimal parameter values.

Because fleet size is the primary objective of the proposed framework, candidate configurations were evaluated using a fleet-first criterion. Let Z_1 denote the smallest fleet size represented in the final archive and let HV_{2D}^{cal} denote the hypervolume of the corresponding nondominated (Z_2, Z_3) set. The calibration score is $F = 10,000 Z_1 - HV_{2D}^{\text{cal}}$, and is minimized. Under the adopted normalization and reference point,

$$0 \leq HV_{2D}^{\text{cal}} \leq 1.05^2 = 1.1025. \quad (53)$$

Consequently, the hypervolume term cannot compensate for even a one-vehicle increase in Z_1 . The calibration criterion therefore preserves strict fleet-size priority while using lower-level Pareto quality to discriminate among configurations attaining the same fleet level.

Adaptive hypervolume normalization.

The hypervolume quantities used for search control and calibration employ adaptive run-specific normalization. For objective j , a raw value $z_j(s)$ is normalized as

$$\tilde{z}_j(s) = \max\left\{0, \frac{z_j(s) - \ell_j}{b_j - \ell_j}\right\}, \quad \ell_j = 0, \quad (54)$$

where b_j denotes the current upper normalization bound. For the selected initial solution s_0 , the initial bounds are

$$b_2^{(0)} = 1.5Z_2(s_0), \quad b_3^{(0)} = 1, \quad b_4^{(0)} = 1.5Z_4(s_0). \quad (55)$$

Whenever a newly observed objective value approaches or exceeds the current normalization range,

$$1.05z_j^{\text{new}} > b_j,$$

the corresponding upper bound is expanded according to

$$b_j \leftarrow 1.2z_j^{\text{new}}. \quad (56)$$

This expansion prevents the normalization scale from becoming inappropriate as the search visits regions outside the initially estimated objective range. For valid Gini

values, $b_3 = 1$ already coincides with the mathematical upper bound and therefore does not require expansion.

Both the two- and three-dimensional hypervolume routines use the normalized reference point $\mathbf{r}^{(m)} = (1.05, \dots, 1.05) \in \mathbb{R}^m$. The margin between 1 and 1.05 allows a solution located at an upper normalization bound to retain a positive hypervolume contribution.

Online hypervolume.

The quantity HV^{online} is used only as an internal search-control signal. During a run, it is repeatedly computed from the nondominated (Z_2, Z_3) solutions at the current minimum fleet level and is used to detect prolonged stagnation.

If one of the adaptive normalization bounds expands, the archive values and the corresponding comparison baseline are renormalized, and the hypervolume-stagnation counter is reset. Although Z_4 remains active in FULL archive dominance, candidate acceptance, archive retention, and final evaluation, an improvement exclusively in Z_4 does not reset the online hypervolume-stagnation counter.

Thus, HV^{online} is a search-management quantity rather than a performance indicator used in the final FULL–No-EQ comparison.

Calibration hypervolume.

After termination of a calibration run, HV_{2D}^{cal} is recomputed from the raw (Z_2, Z_3) vectors at the minimum fleet level using the terminal normalization bounds of that run. This value enters the calibration objective in Eq. (A.1).

Because the terminal bounds depend on the initialization and search trajectory of the individual run, HV_{2D}^{cal} is a run-specific calibration signal and is not suitable for direct cross-variant performance comparison. The common-scale ex-post hypervolumes used to compare FULL and No-EQ are therefore constructed separately in the main computational analysis.

Calibration instances and transfer to No-EQ.

Calibration was conducted on the following seven benchmark instances: c109_21, r101_21, r103_21, r201_21, rc101_21, rc107_21, rc201_21. These instances cover clustered, random, and mixed customer distributions and include both short- and long-horizon benchmark families. They are also included in the final 56-instance benchmark set. Consequently, the final computational assessment is not strictly out of sample with respect to parameter configuration. However, no instance-specific parameter adjustment was performed.

After calibration, the selected FULL configuration was frozen and used throughout the subsequent experiments. No-EQ inherited all common parameter values without independent recalibration. This design ensures that the FULL–No-EQ comparison isolates the system-level effect of removing equity-guided search mechanisms under a shared configuration, rather than introducing separate parameter adaptation as an additional source of variation.

A.2 Calibrated parameters

Table A.1 reports the calibration domains and the selected empirical incumbent. The iteration budget $I_{\max}$ was fixed at 25,000 iterations and was not included in the *irace* search. The remaining 13 parameters governing search-direction transitions, hypervolume stagnation, simulated annealing, adaptive operator control, destroy size, and local-search activation were calibrated.

Table A.1 Calibration domains and empirical incumbent selected by the *irace*-based procedure.

Symbol	Parameter	Calibration domain	Selected
$I_{\max}$	Maximum iterations	{25,000}	25,000
Δ_{seg}	Weight-update segment length	{100, 150, 200, 250}	150
ϵ_{hv}	HV improvement threshold	{0.0005, 0.001, 0.002, 0.005}	0.001
N_{hv}	Maximum stagnant segments	[5, 22]	22
ρ	Operator-weight decay	{0.80, 0.85, 0.90, 0.95}	0.85
s_A	Reward for archive-dominating solutions	[30, 50]	33
s_B	Reward for nondominated solutions	[10, 40]	33
s_C	Reward for dominated solutions	[0, 20]	12
$r_{\min}$	Minimum destroy fraction	[0.05, 0.20]	0.05
$r_{\max}$	Maximum destroy fraction	[0.30, 0.50]	0.40
T_0	Initial SA temperature	{180, 200, 220, 240, 260}	220
α	Cooling rate	{0.993, 0.995, 0.997, 0.999, 0.9992}	0.997
σ	Acceptance scaling factor	{10, 20, 50}	10
λ_{LS}	Local-search activation probability	{0.10, 0.15, 0.20, 0.25}	0.15

Note: N_{hv} , s_A , s_B , and s_C are integer-valued; $r_{\min}$ and $r_{\max}$ are continuous; all other calibrated domains are categorical. The iteration budget $I_{\max}$ was fixed and was not part of the *irace* search.

The selected configuration combines relatively short search-direction segments ($\Delta_{\text{seg}} = 150$) with a conservative stagnation threshold ($N_{hv} = 22$). The operator-update parameters assign the same reward to archive-dominating and newly nondominated solutions ($s_A = s_B = 33$), while dominated but search-useful outcomes retain a smaller positive reward ($s_C = 12$). The calibrated destroy range allows between 5% and 40% of the current customer set to be removed, depending on the selected destroy operation and search state.

The simulated-annealing parameters use an initial temperature $T_0 = 220$, cooling rate $\alpha = 0.997$, and lower-level acceptance scaling factor $\sigma = 10$. Local-search intensification is activated with probability $\lambda_{\text{LS}} = 0.15$. These values are empirical outcomes of the calibration campaign and are not interpreted as universally optimal settings for other instance sets.

A.3 Fixed internal settings

Several internal settings were specified before the calibration campaign and were kept unchanged across all configurations and subsequent experiments. These settings concern the minimum simulated-annealing temperature, local-search termination conditions, beam-search widths, and the temperature level used after perturbation.

Table A.2 reports these values.

Table A.2 Fixed internal settings excluded from the *irace* calibration.

Search-control settings			Beam and perturbation settings		
Symbol	Meaning	Value	Symbol	Meaning	Value
$T_{\min}$	Minimum SA temperature	0.05	W_{out}	Outer beam width	1
I_{LS}	Local-search iteration limit	20	W_{in}	Inner beam width	2
θ_{stop}	Local-search stagnation threshold	7	p_{reheat}	Perturbation reheat target	0.50

The minimum temperature $T_{\min}$ prevents the simulated-annealing temperature from decreasing indefinitely. The local-search controls I_{LS} and θ_{stop} limit the number of intensification iterations and terminate local search after repeated lack of improvement, respectively. The beam widths $W_{\text{out}} = 1$ and $W_{\text{in}} = 2$ determine the number of partial route constructions retained by the beam-search repair procedure. Finally, $p_{\text{reheat}} = 0.50$ controls the temperature restoration applied after stagnation-triggered perturbation.

All calibrated and fixed parameter values reported in Tables A.1 and A.2 were held unchanged throughout the subsequent FULL–No-EQ computational comparison.

B Detailed Algorithms for ALNS Operators

This appendix provides the detailed pseudocodes and summary of the destroy and repair operators utilized in the Multi-Objective ALNS framework.

B.1 Initial solution generation

The initialization procedure combines four deterministic customer-ordering rules with $N_{\text{rand}} = 30$ randomized priority sequences. The construction routine considers the first $W_{\text{scan}} = 15$ customers remaining in the selected order when extending a route.

Algorithm 3: Generate Initial Solutions

Input : Customer set $\mathcal{C}$; number of randomized orders $N_{\text{rand}} = 30$

Output : Set $\mathcal{S}$ of feasible initial solutions

```
1  $\mathcal{S} \leftarrow \emptyset$ ;  
2  $\mathcal{O}_{\text{det}} \leftarrow$   
    $\langle \text{SWEEPORDER}, \text{NEARESTNEIGHBORORDER}, \text{EARLIESTDUEDATEORDER}, \text{TIGHTESTTIMEWINDOWORDER} \rangle$ ;  
  
3 foreach  $o \in \mathcal{O}_{\text{det}}$  do  
4    $s \leftarrow \text{CONSTRUCTSOLUTIONFROMORDER}(o)$ ;  
5   if  $\text{FEASIBLE}(s)$  then  
6      $\mathcal{S} \leftarrow \mathcal{S} \cup \{s\}$ ;  
  
7 for  $i \leftarrow 1$  to  $N_{\text{rand}}$  do  
8    $(w_d, w_{\text{TW}}, w_q) \leftarrow$   
    $(\text{RANDOMUNIFORM}(0, 1), \text{RANDOMUNIFORM}(0, 1), \text{RANDOMUNIFORM}(0, 1))$ ;  
9   foreach  $c \in \mathcal{C}$  do  
10     $\pi(c) \leftarrow w_d \hat{d}_{0c} + w_{\text{TW}} \hat{u}_c^{\text{TW}} + w_q (1 - \hat{q}_c)$ ;  
11    $o_{\text{rand}} \leftarrow \text{SORTASCENDING}(\mathcal{C}, \pi)$ ;  
12    $s \leftarrow \text{CONSTRUCTSOLUTIONFROMORDER}(o_{\text{rand}})$ ;  
13   if  $\text{FEASIBLE}(s)$  then  
14      $\mathcal{S} \leftarrow \mathcal{S} \cup \{s\}$ ;  
  
15 return  $\mathcal{S}$ ;
```

Algorithm 4: Construct Solution from a Customer Order

Input : Ordered list U of unserved customers; scan-window width $W_{\text{scan}} = 15$

Output : Constructed solution s

```
1  $s \leftarrow \emptyset$ ;
2 while  $U \neq \emptyset$  do
3    $r \leftarrow \text{NEWROUTE}(0)$ ;
4    $\text{routeFinished} \leftarrow \text{false}$ ;
5   while not  $\text{routeFinished}$  do
6      $\mathcal{K} \leftarrow \text{FIRST}(\min\{|U|, W_{\text{scan}}\}, U)$ ;
7      $c^* \leftarrow \text{NULL}$ ;
8      $\psi^* \leftarrow +\infty$ ;
9     foreach  $c \in \mathcal{K}$  with rank  $h$  do
10      if not  $\text{FEASIBLEAPPEND}(r, c)$  or not  $\text{RETURNREACHABLE}(r \oplus c)$  then
11        continue;
12       $u_c \leftarrow 1 / \max\{1, \ell_c - \text{CURRENTTIME}(r)\}$ ;
13       $w_c \leftarrow \max\{0, e_c - \text{ARRIVALTIME}(r, c)\}$ ;
14       $\psi_c \leftarrow 0.7d_{\text{last}(r),c} - 30u_c + 0.3w_c + 0.05h d_{\text{last}(r),c}$ ;
15      if  $\psi_c < \psi^*$  then
16         $c^* \leftarrow c$ ;
17         $\psi^* \leftarrow \psi_c$ ;
18      if  $c^* \neq \text{NULL}$  then
19         $r \leftarrow r \oplus c^*$ ;
20         $U \leftarrow U \setminus \{c^*\}$ ;
21      else if not  $\text{DIRECTRETURNFEASIBLE}(r)$  then
22         $v^* \leftarrow \text{BESTRETURNSTATION}(r)$ ;
23        if  $v^* \neq \text{NULL}$  then
24           $r \leftarrow r \oplus v^*$ ;
25           $\text{APPLYREQUIREDPARTIALCHARGE}(r, v^*)$ ;
26        else
27           $\text{routeFinished} \leftarrow \text{true}$ ;
28      else
29         $\text{routeFinished} \leftarrow \text{true}$ ;
30      if  $\text{CONTAINSCUSTOMER}(r)$  then
31         $s \leftarrow s \cup \{\text{CLOSEATDEPOT}(r)\}$ ;
32      else
33        break;
34 if  $U \neq \emptyset$  then
35   foreach  $c \in U$  do
36      $r \leftarrow \text{BESTFEASIBLESINGLECUSTOMERROUTE}(c)$ ;
37     if  $r = \text{NULL}$  then
38       return  $\text{INFEASIBLE}$ ;
39      $s \leftarrow s \cup \{r\}$ ;
40 return  $s$ ;
```

B.2 Destroy Operators

Table B.1 summarizes the destroy operators and the search mechanism targeted by each one. The operators differ primarily in the structure that they attempt to disrupt. Energy-critical removal targets downstream battery risk; route-merging removal targets route consolidation; time-slack removal targets temporal rigidity; fleet-reduction removal directly promotes a reduction in the number of utilized routes; and random removal provides non-directed diversification. The detailed procedures are presented below.

Table B.1 Summary of the destroy operators.

Operator	Main mechanism
Energy-critical removal	Removes customers associated with high downstream energy risk, allowing the repair phase to reconstruct the route and charging plan.
Route-merging removal	Removes customers from structurally compatible routes to encourage their redistribution into fewer routes.
Time-slack removal	Targets customers whose positions strongly restrict downstream schedule flexibility.
Fleet-reduction removal	Empties a selected route and removes additional customers from other routes to create sufficient insertion capacity for route elimination.
Random removal	Removes customers without a directed criterion to diversify the search.

Energy-critical removal.

The first targeted destroy mechanism addresses energy risk that may not be visible from a purely local insertion decision. For every customer on a route, Algorithm 5 compares the available downstream battery with an estimate of the energy needed to continue the route. The resulting risk scores are evaluated relative to the average risk of the corresponding route. Customers whose scores exceed the route-specific threshold form the candidate set from which a removal is selected.

Algorithm 5: Energy Critical Removal Operator

Input : Current solution s ; threshold parameter $\delta = 0.3$ **Output** : Destroyed solution s ; set of unserved customers U $\triangleright$ Phase 1: Initialize candidate set

```
1  $U \leftarrow \emptyset$ ;  
2 for each route  $r \in s$  do  
     $\triangleright$  Phase 2: Calculate energy risk score for each customer on the route  
    3 for each customer  $c \in r$  do  
        4  $next \leftarrow \text{SUCCESSOR}(c, r)$ ;  
        5  $E_c \leftarrow \text{BATTERYLEVEL}(c, r)$ ;  
        6  $\rho(c, r) \leftarrow \max(0, h \cdot d_{c, next} - (E_c - h \cdot d_{c, next}))$ ;  
     $\triangleright$  Phase 3: Calculate average route risk threshold  
    7  $\bar{\rho}_r \leftarrow \frac{1}{|r|} \sum_{c \in r} \rho(c, r)$ ;  
    8  $threshold \leftarrow \bar{\rho}_r \cdot (1 + \delta)$ ;  
     $\triangleright$  Phase 4: Collect candidates exceeding the threshold  
    9  $\mathcal{C} \leftarrow \{c \in r \mid \rho(c, r) > threshold\}$ ;  
     $\triangleright$  Phase 5: Randomly select a candidate and remove it  
    10 if  $\mathcal{C} \neq \emptyset$  then  
        11  $c^* \leftarrow \text{RANDOMSELECT}(\mathcal{C})$ ;  
        12  $s \leftarrow \text{REMOVECUSTOMER}(s, c^*)$ ;  
        13  $U \leftarrow U \cup \{c^*\}$ ;  
        14 if  $|r| = 0$  then  
            15  $s \leftarrow \text{REMOVEROUTE}(s, r)$ ;  
16 return  $(s, U)$ 
```

Algorithm 5 therefore does not attempt to repair the energy deficiency itself. Its role is to expose structurally risky customers to the subsequent repair stage, where both customer positions and charging decisions can be reconsidered.

Route-merging removal.

Whereas Algorithm 5 acts primarily at the customer level, Algorithm 6 operates at the route-structure level. Its purpose is to identify routes that appear sufficiently compatible to justify a joint reconstruction. The procedure first determines how many routes should participate in the removal, identifies potentially mergeable route pairs, and then selects routes according to the adopted spatial-selection strategy. All customers from the selected routes are returned to the unserved set so that the repair procedure can redistribute them without being constrained by their original route boundaries.

Algorithm 6: Route Merging Operator

Input : Current solution s ; number of customers to remove n_{remove}
Output : Destroyed solution s ; set of unserved customers U
▷ Phase 1: Determine target number of routes to remove
1 $m \leftarrow \text{COMPUTETARGETROUTECOUNT}(n_{\text{remove}})$;
▷ Phase 2: Find potential mergeable route pairs
2 $\mathcal{P} \leftarrow \text{FINDPOTENTIALMERGES}(s)$;
▷ Phase 3: Select routes for removal based on strategy
3 **if** $\text{ISCROSSQUADSTRATEGY}()$ **then**
4 | $\mathcal{R}^* \leftarrow \text{SELETROUTESCROSSQUAD}(\mathcal{P}, m)$;
5 **else**
6 | $\mathcal{R}^* \leftarrow \text{SELETROUTESCENROID}(\mathcal{P}, m)$;
▷ Phase 4: Remove routes and extract customers
7 $U \leftarrow \text{EXTRACTCUSTOMERS}(\mathcal{R}^*)$;
8 $s \leftarrow \text{REMOVEROUTES}(s, \mathcal{R}^*)$;
9 **return** (s, U)

The operator deliberately stops after removing the selected route structures. It does not prescribe how their customers must be combined. This separation allows the repair phase to reconstruct one or several routes according to the current feasibility and objective conditions.

Time-slack removal.

The preceding operators target energy and route consolidation. A different source of difficulty arises from customers whose positions consume a large portion of the temporal flexibility of a route. Algorithm 7 therefore begins by computing forward time-slack information and converts this information into customer-specific removal scores. Customers are then removed iteratively, with stochastic selection biased toward those having larger scores. Scores are updated after each removal because changing the route can modify the temporal flexibility of the remaining customers.

Algorithm 7: Time Slack Destroy Operator

Input : Current solution s ; number to remove n_{remove} ; concentration parameter p
Output : Destroyed solution s ; set of unserved customers U
▷ Phase 1: Calculate risk score for each customer
1 $U \leftarrow \emptyset$;
2 $\mathcal{F} \leftarrow \text{COMPUTEFORWARDTIMESLACKS}(s)$;
3 $\mathcal{S} \leftarrow \text{COMPUTEREMOVALSCORES}(s, \mathcal{F})$;
▷ Phase 2: Iterative removal until target is reached
4 **while** $|U| < n_{\text{remove}}$ **do**
5 | $\mathcal{L} \leftarrow \text{SORTBYSORE}(\mathcal{S})$;
6 | $c^* \leftarrow \text{SELECTSTOCHASTICALLY}(\mathcal{L}, p)$;
7 | $U \leftarrow U \cup \{c^*\}$;
8 | $s \leftarrow \text{REMOVECUSTOMER}(s, c^*)$;
9 | $\mathcal{S} \leftarrow \text{UPDATESCORES}(s, \mathcal{S} \setminus \{c^*\})$;
10 **return** (s, U)

Thus, unlike a static ranking followed by the removal of the top n_{remove} customers, Algorithm 7 updates the ranking as the partial solution changes. This makes the removal process responsive to the temporal structure exposed by earlier removals.

Fleet-reduction removal.

The first three targeted operators create partial solutions that may indirectly improve route consolidation. Algorithm 8, in contrast, explicitly attempts to improve the primary objective Z_1 . The procedure considers an ordered collection of increasingly coordinated fleet-reduction neighborhoods. Each neighborhood produces a tentative solution, after which charging quantities, route schedules, feasibility, and all objective values are recomputed before the move can be retained.

Algorithm 8: Fleet-reduction removal

Input : Current solution s
Output : Updated solution s

```

1 if  $|\mathcal{R}(s)| < 2$  then
2   return  $s$ ;
3  $\mathcal{M} \leftarrow (\text{DIRECTROUTEEMPTYING}, \text{EJECTIONCHAIN}, \text{SMALLESTROUTEELIMINATION},$ 
    $\text{MULTIROUTECONSOLIDATION}, \text{THREETOTWOMERGE}, \text{SEGMENTCROSSEXCHANGE}$ 
    $\text{});$ 
4 foreach  $m \in \mathcal{M}$  do
5    $(\hat{s}, \text{success}) \leftarrow m(s)$ ;
6   if  $\text{success}$  then
7     Recompute charging quantities and route schedules in  $\hat{s}$ ;
8     Evaluate  $Z_1(\hat{s})$ ,  $Z_2(\hat{s})$ ,  $Z_3(\hat{s})$ , and  $Z_4(\hat{s})$ ;
9     if  $\text{FEASIBLE}(\hat{s})$  and  $\text{ACCEPTFLEETMOVE}(\hat{s}, s)$  then
10      return  $\hat{s}$ ;
11 return  $s$ ;
```

Algorithm 8 follows a first-success structure: once a feasible fleet-reduction move satisfying the acceptance condition is found, the updated solution is returned. If none of the neighborhoods succeeds, the original solution is preserved. The individual fleet-reduction mechanisms are also used in the local-search fleet-reduction stage described in Appendix C.

Random removal.

Finally, random removal complements the targeted mechanisms above by removing customers without using energy, temporal, consolidation, or objective-specific priority scores. Its purpose is not to address a particular source of infeasibility or inefficiency, but to expose the repair phase to customer combinations that the directed operators may rarely generate. Algorithm 9 summarizes the procedure. A set of customers is sampled uniformly from the current solution, removed from their routes, and returned to the unserved set U for subsequent reconstruction by the selected repair operator.

Algorithm 9: Random Removal Operator

Input : Current solution s ; number of customers to remove n_{remove}
Output : Destroyed solution s ; set of unserved customers U
▷ Phase 1: Collect customers in the current solution

```
1  $U \leftarrow \emptyset$ ;  
2  $\mathcal{C} \leftarrow \emptyset$ ;  
3 foreach route  $r \in s$  do  
4   foreach customer  $c \in r$  do  
5      $\mathcal{C} \leftarrow \mathcal{C} \cup \{c\}$ ;  
6 if  $\mathcal{C} = \emptyset$  then  
7   return  $(s, U)$ ;  
  ▷ Phase 2: Randomly select customers for removal  
8  $q \leftarrow \min\{n_{\text{remove}}, |\mathcal{C}|\}$ ;  
9  $\mathcal{C}^{\text{rem}} \leftarrow \text{UNIFORMRANDOMSAMPLE}(\mathcal{C}, q)$ ;  
  ▷ Phase 3: Remove the selected customers  
10 foreach  $c \in \mathcal{C}^{\text{rem}}$  do  
11    $r \leftarrow \text{ROUTECONTAINING}(s, c)$ ;  
12    $s \leftarrow \text{REMOVECUSTOMER}(s, c)$ ;  
13    $U \leftarrow U \cup \{c\}$ ;  
14   if  $|r| = 0$  then  
15      $s \leftarrow \text{REMOVEROUTE}(s, r)$ ;  
16 return  $(s, U)$ ;
```

B.3 Repair Operators

Table B.2 summarizes the four repair operators and the principal reconstruction mechanism employed by each one. Although all operators ultimately reinsert the customers in the unserved set U , they differ in how they prioritize customers, evaluate insertion positions, and trade immediate insertion quality against future route feasibility. Beam-search repair explicitly retains multiple partial constructions; competitive-station insertion jointly considers direct and charging-assisted alternatives; adaptive insertion changes its insertion criterion according to the current search context; and regret- K repair prioritizes customers for which delaying insertion is potentially costly. The corresponding procedures are detailed below.

Beam-search repair.

Algorithm 10 provides the basic route-construction mechanism used by the repair phase. Rather than committing greedily to a single customer sequence, the procedure retains a restricted set of promising partial route constructions. This is particularly useful in the electric vehicle setting because an insertion that appears attractive from a routing perspective may later require an unfavorable charging detour or may reduce the feasibility of subsequent insertions.

The procedure operates at two nested levels. The inner level expands partial customer sequences, whereas the outer level retains the most promising alternatives after charging requirements and station-placement implications have been considered. The

Table B.2 Summary of the repair operators.

Operator	Main mechanism
Beam-search repair	Jointly explores customer sequencing and charging-station placement while retaining a restricted set of promising partial routes.
Competitive-station insertion	Evaluates direct and station-assisted insertion alternatives concurrently and selects the best feasible option.
Adaptive insertion	Changes its insertion criterion according to the current fleet-reduction state and objective emphasis.
Regret- K repair	Prioritizes customers for which delaying insertion is expected to cause the largest deterioration in future insertion quality.

two beam widths W_{in} and W_{out} control how many candidate states are preserved at the corresponding levels.

Algorithm 10: Beam Search Repair Operator

Input : Current solution s ; set of unserved customers U ; beam widths $W_{\text{out}}, W_{\text{in}}$
Output : Repaired solution s

```

1 while  $U \neq \emptyset$  do
2    $c^* \leftarrow \text{SELECTSEEDCUSTOMER}(U)$ ;
    $\triangleright$  Initialize new route from a seed customer
3    $\mathcal{B}_{\text{out}} \leftarrow \text{INITBEAMSTATES}(c^*)$ ;
    $\triangleright$  Two-tier beam search (Subroutine)
4   while  $\neg \text{TERMINATIONCONDITION}(\mathcal{B}_{\text{out}})$  do
    $\triangleright$  Inner tier: expand customer sequences
5      $\mathcal{B}_{\text{in}} \leftarrow \text{EXPANDINNERBRANCHES}(\mathcal{B}_{\text{out}}, U)$ ;
6      $\mathcal{B}_{\text{in}} \leftarrow \text{TOPK}(\mathcal{B}_{\text{in}}, W_{\text{in}})$ ;
    $\triangleright$  Outer tier: select the best charging station placement
7      $\mathcal{B}_{\text{out}} \leftarrow \text{TOPK}(\mathcal{B}_{\text{in}}, W_{\text{out}})$ ;
    $\triangleright$  Finalize route
8      $r^* \leftarrow \text{EXTRACTBESTROUTE}(\mathcal{B}_{\text{out}})$ ;
9      $s \leftarrow s \cup \{r^*\}$ ;
10     $U \leftarrow U \setminus \text{ASSIGNEDCUSTOMERS}(r^*)$ ;
11 return  $s$ 
```

The operations in Lines 2–8 define the subroutine `BEAMSEARCHROUTEBUILD`(c^*, U), which constructs one route starting from seed customer c^* . The outer loop repeatedly invokes this subroutine until all customers in U have been assigned, thereby defining the complete `BEAMSEARCHREPAIR` operator.

This distinction is also useful for interpreting the remaining repair operators. Competitive-station insertion, adaptive insertion, and regret- K repair primarily attempt to insert customers into the existing route set; when this is no longer possible, they can call `BEAMSEARCHROUTEBUILD` as a common fallback mechanism for constructing a new feasible route.

Competitive-station insertion.

Algorithm 11 focuses on the interaction between customer insertion and charging-station placement. A sequential policy that first tests direct insertion and considers a charging station only after direct insertion fails may overlook a station-assisted alternative with a better overall route impact. The competitive-station operator therefore constructs both sets of alternatives before selecting the insertion to apply.

For a selected customer c , the procedure evaluates feasible direct positions and charging-station-assisted positions concurrently. If neither set contains a feasible insertion, an extended station search is attempted. Only when these alternatives fail does the operator create a new route using the beam-search subroutine defined in Algorithm 10.

Algorithm 11: Competitive Station Insertion Operator

Input : Set of unserved customers U ; current solution s
Output : Repaired solution s

```

1 while  $U \neq \emptyset$  do
2    $c \leftarrow \text{SELECTNEXTCUSTOMER}(U)$ ;
3    $\triangleright$  Evaluate both strategies concurrently
4    $P_{\text{direct}} \leftarrow \text{EVALDIRECTINSERTIONS}(c, s)$ ;
5    $P_{\text{station}} \leftarrow \text{EVALSTATIONASSISTEDINSERTIONS}(c, s)$ ;
6    $p^* \leftarrow \text{SELECTBESTFEASIBLE}(P_{\text{direct}} \cup P_{\text{station}})$ ;
7   if  $\neg \text{ISFEASIBLE}(p^*)$  then
8      $p^* \leftarrow \text{EVALEXTENDEDSTATIONSEARCH}(c, s)$ ;
9   if  $\text{ISFEASIBLE}(p^*)$  then
10     $s \leftarrow \text{APPLYINSERTION}(s, p^*)$ ;
11     $U \leftarrow U \setminus \{c\}$ ;
12  else
13     $r_{\text{new}} \leftarrow \text{BEAMSEARCHROUTEBUILD}(c, U)$ ;
14     $s \leftarrow s \cup \{r_{\text{new}}\}$ ;
15     $U \leftarrow U \setminus \text{ASSIGNEDCUSTOMERS}(r_{\text{new}})$ ;
16 return  $s$ 

```

Algorithm 11 therefore treats charging support as part of the customer-insertion decision itself rather than as a correction applied after a customer has already been positioned. The result is a reconstruction mechanism in which route position and charging support can compete on the same insertion decision.

Adaptive insertion.

Whereas Algorithm 11 changes the set of insertion alternatives, Algorithm 12 changes the criterion by which alternatives are evaluated. The active insertion measure is selected from the current search state and the lower-level objective-weight vector $\mathbf{w} = (w_d, w_g, w_t)$. This allows the same reconstruction mechanism to emphasize different search goals without requiring a separate repair operator for each lower-level objective.

When the search is in fleet-reduction mode, the operator emphasizes vehicle packing. Otherwise, it selects among distance, active-workload, and maximum-completion-time criteria according to the current weight vector. Customers are processed using a

tight-time-window priority, after which the operator first attempts insertion according to the selected measure and then uses charging-assisted insertion or beam-search route construction when necessary.

Algorithm 12: Adaptive Insertion Operator

Input : Current solution s ; set of unserved customers U ; objective weights $\mathbf{w} = (w_d, w_g, w_t)$; flag $vrMode$

Output : Updated solution s

▷ Phase 1: Determine contextual insertion cost function

```

1 if  $vrMode = \text{TRUE}$  then
2   |  $f \leftarrow \text{VEHICLEPACKING}$ ;
3 else if  $w_d > 0.6$  then
4   |  $f \leftarrow \text{DISTANCE}$ ;
5 else if  $w_g > w_t$  then
6   |  $f \leftarrow \text{WORKLOAD}$ ;
7 else
8   |  $f \leftarrow \text{MAXTIME}$ ;
9   | ▷  $\text{MAXTIME}$  minimizes  $Z_4$ : maximum route completion time
10  |
11  | ▷ Phase 2: Insert based on tightest time window priority
12 while  $U \neq \emptyset$  do
13   |  $c \leftarrow \text{SELECTTIGHTESTTIMEWINDOW}(U)$ ;
14   |  $p^* \leftarrow \text{FINDBESTINSERTION}(c, s, f)$ ;
15   | if  $\text{ISFEASIBLE}(p^*)$  then
16   |   |  $s \leftarrow \text{APPLYINSERTION}(s, p^*)$ ;
17   |   |  $U \leftarrow U \setminus \{c\}$ ;
18   | else
19   |   | ▷ Attempt station-assisted insertion
20   |   |  $p^* \leftarrow \text{COMPETITIVESTATIONINSERT}(c, s)$ ;
21   |   | if  $\text{ISFEASIBLE}(p^*)$  then
22   |   |   |  $s \leftarrow \text{APPLYINSERTION}(s, p^*)$ ;
23   |   |   |  $U \leftarrow U \setminus \{c\}$ ;
24   |   | else
25   |   |   | ▷ Fallback to route initialization
26   |   |   |  $r_{\text{new}} \leftarrow \text{BEAMSEARCHROUTEBUILD}(c, U)$ ;
27   |   |   |  $s \leftarrow s \cup \{r_{\text{new}}\}$ ;
28   |   |   |  $U \leftarrow U \setminus \text{ASSIGNEDCUSTOMERS}(r_{\text{new}})$ ;
29 return  $s$ 

```

The procedure therefore links the repair phase directly to the current multi-objective search direction. Importantly, changing f affects the ranking of insertion alternatives but does not relax feasibility requirements. Every retained insertion is still subjected to the route and partial-recharging feasibility evaluation described in the main methodology.

Regret- K repair.

The preceding operators primarily determine where a selected customer should be inserted. Algorithm 13 additionally addresses the question of which customer should be

processed first. The underlying idea is that customers with few attractive alternatives should be handled before those having several comparable insertion possibilities.

For every unserved customer, the operator computes up to K feasible insertion alternatives ranked by their raw insertion cost. Its regret value measures how much worse the alternative positions are relative to the best one. The customer with the largest regret is selected first. After customer selection, packing rewards may reorder its insertion positions to favor better utilization of existing routes.

Algorithm 13: Regret- K Repair Operator

Input : Set of unserved customers U ; current solution s ; regret depth $\bar{K}_{\text{regret}}$
Output : Repaired solution s

```

1 while  $U \neq \emptyset$  do
    ▷ Phase 1: Calculate opportunity cost for each customer
2    $K \leftarrow \text{ADJUSTREGREDEPTH}(U, \bar{K}_{\text{regret}})$ ;
3   for each  $c \in U$  do
4      $L_c^{\text{raw}} \leftarrow \text{FINDKBESTINSERTIONS}(c, s, K)$ ;
5     ▷ Sorted ascending by raw distance cost
6      $R(c) \leftarrow \sum_{i=2}^K (L_c^{\text{raw}}[i].\text{cost} - L_c^{\text{raw}}[1].\text{cost})$ ;
7     ▷ Regret computed on raw costs before packing adjustment
8      $L_c^{\text{ranked}} \leftarrow \text{APPLYPACKINGREWARDS}(L_c^{\text{raw}}, s)$ ;
9     ▷ Packing reward reorders list for insertion-point selection only
10    ▷ Phase 2: Insert customer with highest regret
11     $c^* \leftarrow \arg \max_{c \in U} R(c)$ ;
12     $p^* \leftarrow L_{c^*}^{\text{ranked}}[1]$ ;
13    ▷ Uses packing-adjusted ordering
14    if  $\text{ISFEASIBLE}(p^*)$  then
15       $s \leftarrow \text{APPLYINSERTION}(s, p^*)$ ;
16       $U \leftarrow U \setminus \{c^*\}$ ;
17    else
18       $r_{\text{new}} \leftarrow \text{BEAMSEARCHROUTEBUILD}(c^*, U)$ ;
19       $s \leftarrow s \cup \{r_{\text{new}}\}$ ;
20       $U \leftarrow U \setminus \text{ASSIGNEDCUSTOMERS}(r_{\text{new}})$ ;
21  return  $s$ 

```

Two ranking stages should be distinguished in Algorithm 13. The regret value $R(c)$, which determines which customer is processed next, is computed from the raw insertion costs. The packing adjustment is applied only afterward to choose the insertion position for the selected customer. Thus, vehicle-packing considerations do not alter the underlying regret-based customer priority.

As in the other repair operators, failure to identify a feasible insertion into the existing route set triggers `BEAMSEARCHROUTEBUILD`. The four repair mechanisms therefore share a common feasibility-preserving fallback while differing in how they explore customer order, insertion position, charging support, and objective-specific search emphasis.

C Local Search

This appendix details the fleet-reduction component used by the local-search procedure described in Section 4.7. While the routing VND primarily improves customer sequencing and assignment at a fixed fleet size, reducing the number of utilized vehicles generally requires more coordinated modifications. Eliminating one route may require several of its customers to be redistributed simultaneously, and these changes can in turn alter charging requirements, route schedules, and the feasibility of neighboring routes.

Algorithm 14 therefore applies a collection of fleet-reduction neighborhoods in an ordered sequence. The neighborhoods range from relatively direct route-elimination attempts to increasingly coordinated redistribution mechanisms. The procedure begins with direct route emptying and ejection-based redistribution, and subsequently considers charging-aware elimination, pairwise consolidation, three-to-two merging, and segment-based redistribution. This ordering allows comparatively simple fleet-reduction opportunities to be exploited before more disruptive restructuring is attempted.

Each neighborhood m receives the current feasible solution s and returns a tentative solution $\hat{s}$ together with a success indicator. A successful structural move is not accepted immediately. Because customer redistribution can change energy consumption, charging-station requirements, and arrival times, all affected routes are re-evaluated before the move is retained. In particular, partial-charging quantities and route schedules are recomputed and feasibility is checked under the time-window, payload, and battery constraints.

Algorithm 14: Local-Search Fleet-Reduction Procedure

Input : Current feasible solution s
Output : Updated feasible solution s

```
1 if  $|\mathcal{R}(s)| < 2$  then
2   return  $s$ ;
   // Construct the ordered fleet-reduction neighborhood list
3  $\mathcal{M} \leftarrow \langle \rangle$ ;
4 Append DIRECTROUTEEMPTYING to  $\mathcal{M}$ ;
5 Append EJECTIONCHAINREDISTRIBUTION to  $\mathcal{M}$ ;
6 Append CHARGINGAWAREROUTEELIMINATION to  $\mathcal{M}$ ;
7 Append PAIRWISEROUTECONSOLIDATION to  $\mathcal{M}$ ;
8 Append THREETOTWOROUTEMERGE to  $\mathcal{M}$ ;
9 Append SEGMENTBASEDREDISTRIBUTION to  $\mathcal{M}$ ;
   // Apply the neighborhoods under a first-success policy
10 foreach  $m \in \mathcal{M}$  do
11    $(\hat{s}, success) \leftarrow m(s)$ ;
12   if  $success$  then
13     Recompute partial-charging quantities and route schedules in  $\hat{s}$ ;
14     Evaluate all modified routes in  $\hat{s}$ ;
15     if FEASIBLE( $\hat{s}$ ) and ACCEPT $_m(\hat{s}, s)$  then
16       Fully evaluate  $\hat{s}$  with respect to  $(Z_1, Z_2, Z_3, Z_4)$ ;
17       return  $\hat{s}$ ;
18   Roll back all tentative modifications made by  $m$ ;
19 return  $s$ ;
```

Algorithm 14 follows a first-success policy. Once a neighborhood generates a feasible move satisfying its acceptance condition, the resulting solution is fully evaluated with respect to (Z_1, Z_2, Z_3, Z_4) and immediately returned to the main local-search procedure. The remaining neighborhoods are therefore examined only when the preceding mechanisms fail to produce an acceptable fleet-reduction move.

The rollback operation is also important. A neighborhood may successfully construct a tentative redistribution while the resulting routes become infeasible after charging and schedule propagation. In such a case, all temporary modifications are discarded before the next neighborhood is tested. Consequently, every neighborhood starts from the same feasible incumbent solution rather than from an infeasible intermediate state.

The neighborhoods differ in the degree of coordination used to eliminate a route. DIRECTROUTEEMPTYING attempts to reinsert the customers of a target route directly into the remaining routes. EJECTIONCHAINREDISTRIBUTION allows an insertion to displace other customers and propagates these displacements across routes when a direct emptying is not possible. CHARGINGAWAREROUTEELIMINATION additionally accounts for the charging changes induced by customer reassignment. PAIRWISEROUTECONSOLIDATION restructures two routes jointly, whereas THREETOTWOROUTEMERGE attempts a larger consolidation in which the customers of three routes are reassigned to two routes. Finally, SEGMENTBASEDREDISTRIBUTION

exchanges or transfers consecutive customer segments to create additional capacity and temporal flexibility for route elimination.

These mechanisms complement the fleet-reduction destroy operator in Algorithm 8. The destroy operator creates large reconstruction opportunities within the ALNS destroy–repair cycle, whereas Algorithm 14 acts as an intensification procedure on an already feasible solution. In both cases, any retained fleet change is subject to complete route, charging, and objective re-evaluation before it can affect the current solution or Pareto archive.